

Idioms, Polysemy, and Lexical Collocations Across AI Models

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TL; DR

1. Analysis of 127 LLMs (111 open-weight + 16 commercial) on the “La Ghigliottina” (“The Guillotine”) game, adapted into English with an original dataset of 30 items.
2. No model systematically matches the author’s solution; the best open-weight model (stepfun/step-3.5-flash, 71.1%) and the best commercial model (x-ai/grok-4.6, 76.7%) remain far from 100% and are not statistically distinguishable from the other top models in their own group.
3. Including manually validated synonyms, extended correctness rises to 80.0% (bytedance-seed/seed-2-1-turbo): part of the distance from the solution is a limitation of the automatic metrics, not a genuine failure of the model.
4. On at least three items (supermarket/checkout, emotion/stress, watching/watch), the models’ convergent answer turned out to be more coherent than the author’s original solution.
5. Temperature (0.0/0.3/0.7) has no significant aggregate effect on accuracy; the models’ ranking stays stable across temperatures.
6. Model size correlates strongly with accuracy ($\rho=0.756/0.690$) but is not the only factor: gpt-oss:20b (only 3.61B active parameters, MoE) remains among the best.
7. Lexical rarity (Zipf) does not explain difficulty in general ($\rho=0.246$, not significant); it remains a real factor only for the isolated case of petrichor.
8. guillo_10 (PC) is the single hardest item overall (0.000 out of 193 attempts). No model chooses the acronym, even while explicitly noting it in its reasoning.
9. starcoder2:15b-instruct, in 46/90 responses, answers correctly and then keeps generating unrelated Python code until it hits the token limit.
10. deepseek-r1:14b repeatedly oscillates between MOUSE (correct) and BEAST, arbitrarily settling on BEAST. Correct propositional knowledge, but no procedural control to stop (Ryle’s Knowing-That vs Knowing-How).
11. Cohen’s Kappa between the automatic metrics and the manual synonym judgment is low (-0.041). The full manual review recovered 468 valid synonyms not captured by the automatic metrics.
12. The three difficulty measures (author’s label, accuracy cluster, time cluster) agree only weakly with each other (weighted Kappa 0.122-0.325). Difficulty as perceived by the model and difficulty declared a priori are related but not equivalent constructs.
13. RQ6 (the author’s qualitative judgment): Claude Opus 5 is the chosen model, with a bias openly declared by the author, for the quality of its reasoning process rather than for raw accuracy.
14. Main limitations: N=1 on the reference judgment, non-uniform sampling parameters for commercial models, reproducibility not guaranteed over time (new releases/quantizations/backends).
15. Future work: mechanistic interpretability/XAI, widening the temperature range, submitting the dataset to human review with N>30.

Abstract

“La Ghigliottina” (“The Guillotine”) is an Italian television game of semantic association: five clue words must lead to a single solution word through idioms, polysemy, and lexical collocations. This study transposes the game into English and puts it to 127 large language models (LLMs) — 111 open-weight and 16 commercial — to investigate whether and how models apply this kind of associative reasoning. An original dataset of 30 items in English was built, submitted to each open-weight model at three temperatures (0.0/0.3/0.7) and to the commercial models with a single run, for a total of 9,380 valid observations on the open-weight models and 444 on the commercial ones.

The results answer six research questions. No model systematically matches the reference solution: the best open-weight model (*stepfun/step-3.5-flash*, 71.1% exact matches) and the best commercial model (*x-ai/grok-4.6*, 76.7%) remain far from perfect, and pairwise comparison (McNemar with FDR correction) finds no statistically distinguishable winner within either top group. Including manually validated synonyms, extended correctness rises to 80.0%, signalling that part of the measured distance is a limitation of the automatic metrics rather than a semantic failure of the models. On at least three items, the models’ convergent answer (checkout, stress, watch) turns out to be more coherent than the solution originally chosen by the author. Temperature has no significant aggregate effect on accuracy or on the models’ ranking, while model size correlates strongly with accuracy (Spearman rho up to 0.756) without fully explaining it, as shown by the case of *gpt-oss.20b*, competitive despite only 3.61 billion active parameters per token thanks to its Mixture-of-Experts architecture. The question of which model performs best according to the author’s judgment (RQ6) does not admit a purely statistical answer: *Claude Opus 5* is proposed, motivated by the procedural quality of the reasoning observed rather than by aggregate accuracy, with a personal bias explicitly declared.

Beyond the aggregate metrics, the qualitative analysis of the reasoning logs reveals specific failure patterns. The single hardest item overall (PC, 0.000 accuracy out of 193 attempts) shows how models notice the correct clue (the acronym) and discard it anyway in favour of more natural forms such as COMPUTER. The model *starcoder2:15b-instruct*, in over half of its responses, answers correctly and then keeps generating Python code entirely unrelated to the task until it hits the token limit. *deepseek-r1:14b* repeatedly oscillates between the correct answer (MOUSE) and a wrong alternative (BEAST) until arbitrarily settling on the wrong answer, despite having explicitly stated “*I give up; the answer is MOUSE*” shortly before. These episodes are read through Ryle’s distinction between Knowing-That and Knowing-How: the models possess the correct propositional knowledge but lack the procedural control to stop on it. Lexical rarity (Zipf analysis) does not explain the dataset’s difficulty as a whole (rho = 0.246, not significant), but remains a plausible factor for the isolated case of petrichor, the only genuinely rare solution word in the corpus. The three difficulty measures used (the author’s declared label, the accuracy cluster, the generation-time cluster) agree only weakly with one another (weighted Kappa 0.122-0.325), indicating that the difficulty perceived by the models and the difficulty assigned a priori by the author are related but not equivalent constructs.

The study has declared limitations: the reference judgment (solution, difficulty, coherence) is the work of a single author (N=1). Sampling parameters are not uniform across commercial models due to structural constraints of the interfaces. The reproducibility of the experiment is not guaranteed over time given the continuous release of new models, updates to existing ones, and changes in quantization. Future work includes the use of mechanistic interpretability techniques, widening the temperature range, and submitting the dataset to human review on a larger sample (N>30) for a statistically comparable comparison with model performance.

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Introduction

“La Ghigliottina” is a game that is part of a television programme broadcast on the Italian national network called “L’Eredità” (“The Inheritance”) (names, terms, and conditions are the property of RAI, Italian public broadcasting). The host provides five pairs of terms. On each turn the contestant must choose one of the pair. Choosing the correct word means it is selected as a clue; choosing wrongly halves the prize pot. From the second pair of words onward, the contestant can make choices or reason it through. The final goal is to have five clue words useful for finding a common word (the answer is chosen in advance and deposited with a notary, guaranteeing its authenticity and uniqueness) as the solution. The contestant always learns the correct clues. After due time for reflection, the contestant reveals the word, explaining to the host and the audience what associations led them to the final word. If we had to summarize the goal of the game in one question, it would be: which word connects the five clues, considered as idioms, concepts, or connections? If the contestant’s word matches the one deposited, the prize sum is paid out; otherwise, they can return in the next episode for a second chance. In the episode of March 11, 2025, the following words were given: carta (paper), legno (wood), sogno (dream), effetto (effect), scarpe (shoes). The contestant’s answer was: scarpe (shoes). The correct answer was: lucido (literally “shiny/polish”). The correct association: carta da lucido (tracing paper), lucido per legno (wood polish), sogno lucido (lucid dream), effetto lucido (glossy effect), and lucido per scarpe (shoe polish). In English translation, the association might not jump out immediately, due to linguistic and semantic differences.

From the research carried out, this game does not exist anywhere else in the world. There is a psychological test called the Remote Associates Test (RAT — Olteteanu AM & Hassan Zunjani 2020), a test used to measure creative potential. Unlike the game, the RAT has only three words for which a fourth must be found. One example is the association of these three words: bass, complex, sleep, with solution deep. To an English-speaking listener the association would jump out immediately; for a native Italian speaker this association is fairly obscure. In the same way, the English translation of the Italian game would remain obscure to an English speaker. The RAT consists of 30 items to be completed in 40 minutes.

To replicate the game in English, it is not enough to translate the Italian version; it is necessary to create 30 new items starting from English semantics and linguistics, so as not to run into translation issues that would cause the sense of the answer to be lost. For example, translating the previous five words literally would give these five clues: paper, wood, dream, effect, shoes. The word “Lucido” could be translated as: shiny. An English speaker might make different associations, for example: Paper – House of paper (or Paper house); Wood – Wooden house / House made of wood; Dream – Dream house; Effect – House effect (as in the greenhouse effect, or the visual/scenic impact in architecture and theatre); Shoes – House shoes (slippers/indoor footwear), arriving at the final word HOUSE. In semantic-association games based on idioms, polysemy, and lexical collocations, the language in which the clues are given is fundamental to obtaining a coherent answer.

LLMs of every size have been trained using a fairly large vocabulary. For AI models we cannot speak of words, but of tokens. For example, *Llama 3.2 1B* and *3B* have a vocabulary of 128K tokens, *Qwen 2.5 1.5B* 151K tokens, while *Gemma 2 2B* has 256K tokens. Even large-scale models such as *GPT4* or the *Gemini* family have 200K and 256K tokens respectively. Complex or long words, or words in a language other than the one used in pre-training, can be split into smaller pieces and sometimes lose their meaning. Models can be pre-trained as monolingual or multilingual in order to be able to interact

across multiple countries. English is the language on which all models, in varying proportions, are pre-trained.

What would happen if we submitted an English version of the guillotine game to AI models of various sizes?

In this paper we will analyse and try to answer the following questions:

1. Does the author’s solution match the one found by the models?
2. Quantitatively, how far are the models’ answers from the reference solution?
3. In the word association, can a model’s answer turn out to be more coherent than the author’s (subjective linguistic judgement)?
4. How much does a model’s answer change across the temperature spectrum, at equal top_k/top_p?
5. Does model size (parameters) correlate with the accuracy of the association?
6. Which model gives the best answer, according to the author’s judgement?

The analysis lasted about 20 days, ending on August 22, 2026. The continuous release of new models, updates to existing ones, different quantizations, or inference backends do not guarantee full reproducibility of the experiment.

Method

Various LLMs of different sizes and reasoning capabilities were selected in order to widen the number of observations. In total, 127 LLMs were used, ranging from 350M parameters to models with trillions of parameters. Many models do not disclose their total parameter count, making it impossible to identify a precise number. Two types of computing services were used: models downloadable from Ollama (<https://ollama.com/>); an online processing service (<https://openrouter.ai/>) with the purchase of credits.

Processing was automated by creating custom Python scripts. Open-weight models were processed using the Ollama provider; for the online models, the service's dedicated API was used after subscribing. Provisional data were saved to an Excel file. Additional data provided by the online service were saved to a JSONL file for further analysis. Regardless of the provider on which they were processed, they will be referred to as open-weight.

The time allowed for the contestant to give an answer ranges from 30 to 60 seconds (depending on the version of the game); open-weight reasoning models were given a limit of 600 seconds to finish reasoning. This time was chosen based on a few preliminary, empirical trials. On average, to complete a reasoning process within the available token limit, the time ranged between 420 and 540 seconds (7–9 minutes). Some models took over 1800 seconds to complete their reasoning without producing a valid answer (empty string). The choice of 600 seconds reflects both the observed reasoning timings and an acceptable limit for models to be able to generate a valid answer. In this experiment we set the time tenfold higher than the contestant's, to be sure of obtaining an answer. From this, we can ask another question: if the contestant were given ten minutes to identify the correct word, would they be able to provide the right one?

For the remaining models, this was not a configurable parameter.

The extracted data were saved to an Excel file and subsequently processed with a Jupyter notebook.

Dataset

A dataset of 30 items was created, each containing five clue words and one answer, entirely in English. The meaning of the answer was made explicit in the original file. The words were created directly by the author, without external help. To provide a summary assessment of the items created, a first review was carried out with the support of an AI LLM (*Claude Sonnet 5*). Following this first review, the author modified 5 items to bring them into the optimal category. For example, one of the replaced items:

Words: capital; arteriocardial; recapitulation; escapism; decapitated

Common word: capi (“capi” is Italian for “heads/bosses/garments”)

Difficulty: hard

Explanation: all words have “capi” inside

The five words do not share an association through a common element; it is not a semantic association or an idiom, only a spelling pattern. This type of item does not test associative reasoning, but the search for grammatical differences/similarities.

The review carried out by the AI LLM is not a validation. Only by submitting the 30 items to a formal review can we obtain a review and confirm the correctness of the common word (as in the RAT).

The 30 items were submitted to all models according to the following methodological decisions

Decision	Choice	Rationale
Prompt	Single, fixed, in English, the same for all models	Isolates the model as the independent variable. Customizing the prompt per model would introduce a confounding factor (it would measure prompt-engineering quality, not model capability)
Temperature	0.0 / 0.3 / 0.7	Covers the spectrum from deterministic to moderately stochastic. At T=0.0, top_k/top_p are inert (greedy decoding)
top_k / top_p	40 / 0.9, fixed across all open-weight runs	Continuity with the parameters of previous projects
Automatic metrics	exact_match, same_root (Porter stemmer, nltk), levenshtein_normalized (rapidfuzz, similarity threshold 0.5)	Case-insensitive comparison. same_root captures regular morphological variation (not irregular forms, but mitigated by manual review). Automates a semantic part that is easily identified through linguistic patterns
Manual metrics	synonym_manual, coherence_manual	Filled in by the author after data collection, not reliably automatable
Commercial models	Single run per model (not three temperatures), sampling parameters not controlled	Web interfaces do not expose temperature/top_k/top_p; even via API, different providers expose different parameters (e.g., the OpenAI API has no top_k). Treated as a separate track, not comparable 1:1 with the open-weight models
Output	An Excel file with 4 sheets (T_0.0 / T_0.3 / T_0.7 / Commercial), long/tidy format. A JSONL file for additional information	One row = one generation with its various parameters. Storage of parameters useful for in-depth analysis

Table 1 Methodological decisions applied

Preliminary Assessment

Each item was classified according to an arbitrary difficulty level set by the author. The difficulty of finding the common word may differ from person to person. In total, a corpus of 30 items is proposed. At the time of writing, no review by human subjects had been carried out. Although it is not part of the research questions, it would be interesting to discover the difference in answers between AI and humans. To this end, the reader is invited to play the game. In the table below, the five clues given to the AI models are proposed. The reader is invited to fill in the two columns: “Solution” and “Difficulty”. In the “solution” column, the word believed to be the common and correct one for the five clues is entered (one, and only one, word). In the “difficulty” field, the perceived difficulty in identifying the correct word is entered. There are 3 difficulty levels: 1 = easy; 2 = medium; 3 = hard. These are the same difficulty levels used by the author to set the initial parameter. The time available for filling this in should be about 30 minutes. Readers may take whatever time they prefer; we ask that the total time, or the time per item, be recorded in order to allow a comparison with the various LLMs. In Appendix I, the solution, the difficulty level, and the reason the solution word was assigned.

word1	word2	word3	word4	word5	solution	Difficulty
gluteus	adjustable	sit	office	kitchen		
highway	wheel	transport	fuel	insurance		
Washington	skyscraper	sewer	courthouse	shop		
Pascal	inch	stress	cloud	blood		
moon	speed	lamp	hope	visible		
videogame	human	society	advance	history		
Patrick	sun	wars	five	multiplication		
card	dental	suspension	nose	burn		
rain	arid	perfume	terrain	lift		
CPU	RAM	GPU	HD	VRAM		
language	snake	colt	monty	biceps		
qualification	book	academic	noble	job		
gestational	politics	court	another	substitute		
geometric	meditation	art	powder	sanskrit		
mean	math	analysis	descriptive	inferential		
aircraft	geometry	woodworking	thought	surface		
cloth	rectangular	pole	symbol	signal		
game	television	bird	observe	vigilante		
double	red	fingers	angry	necklace		
exotic	adventure	fly	culture	relax		
exit	shelves	cart	pay	customer		
reaction	response	driven	cognitive	adrenaline		
philosophy	math	physics	business	society		
construct	lamp	problem	innovation	good		
tree	pie	computer	day	York		
love	Plato	essentialism	taoism	knowledge		
renewable	resource	environmental	social	economic		
countryside	hole	whiskers	Mickey	lab		
barrel	web	wood	rusty	cat		
Today	simple	gift	time	now		

Table 2 The 30 items of the evaluation protocol to be submitted to human subjects for review

Readers are invited to share their solutions in order to have a larger basis for assessing difficulty and the solutions found. With a sufficient number of reader responses, it is possible to compare the responses with the LLMs.

Prompt

The prompt was generated based on the guidelines of Ronanki et al. (2025) and used for all open-weight models, so as not to alter the prompt variable.

“You are playing a word-association game. You will be given 5 clue words. Find the single word that connects all five (through common phrases, idioms, or shared concepts). Respond with ONLY the final word, in uppercase, with no explanation. Clues: {word1}, {word2}, {word3}, {word4}, {word5}.”

In the Python file, a function was created to isolate the answer from the reasoning.

Human Evaluation

At the end of processing, the Excel file was reviewed by the author (N=1 reviewer). The review involved entering the value 1 in the `synonym_manual` and `coherence_manual` columns. `synonym_manual` is filled in if the word given is a synonym of the target word — for example, “travel” can have synonyms such as “journey” or “voyage”. `coherence_manual` is filled in if the word given has a reasonably correct connection compared to the target word. These columns could not be filled in automatically. Human review was necessary to validate the outputs of the various models. One of the research questions (RQ3) sets out to understand whether models provide a more coherent word than the one set; analysing the answers and validating them as verified alternatives was necessary.

Analysis

At the end of the author’s review, Python scripts and notebooks were created for data analysis.

In total, 111 open-weight models were used, with 90 records expected for each (30 items $\times$ 3 temperatures). Only one model turned out incomplete (*arcee-ai/virtuoso-large*, 8/90 rows). Overall, 270 duplicate rows (same model/item/temperature, present in multiple source files) were removed, keeping the first occurrence. After filtering on the status column (status = “ok” excludes timeouts and errors), 9,380 valid observations remain on the open-weight models and 444 valid observations on the 16 commercial models (out of 480 total; 36 errors, almost all concentrated on *meta/muse-spark-1.2*, which failed all 30 calls, and on *inception/mercury-2*, with 6 errors out of 30). 108 open-weight models have sufficient data for the accuracy ranking. The 16 commercial models were collected without temperature parameters, obtaining 30 observations.

Correctness Metrics

Three automatic metrics per row: `exact_match` (exact equality with the expected solution), `same_root` (same morphological root), `is_similar_lev` (normalized Levenshtein similarity above threshold). Added to these are two manually filled columns, with review completed 100% on both datasets: `synonym_manual` (the answer is a valid synonym not caught by the automatic metrics) and `coherence_manual` (linguistic coherence judgement). The `is_clue_echo` column (the model repeats one of the 5 clue words instead of finding the connector) and `format_compliant` are not correctness measures but help interpret it. A model with low `exact_match` and high `is_clue_echo` is failing differently from one with low `is_clue_echo`.

Statistical Choices

- **Wilson confidence intervals** (not Wald/normal) for every accuracy rate, because they are more reliable with proportions near 0 or 1 and with samples that are not huge.
- **Spearman correlation** (not Pearson) for RQ5 (size vs. accuracy): what matters is whether larger size and accuracy ranking move together, not the existence of a linear relationship.
- **Cochran’s Q** (not a chi-square test) for RQ4: the same 30 items are presented again to the same model at three temperatures — a repeated-measures design, not independent samples.
- **McNemar + FDR correction (Benjamini-Hochberg)** for pairwise comparison among the top models. The data are paired, and with many parallel comparisons the risk of false positives grows. FDR correction is necessary, not optional.
- **Cohen’s Kappa** to quantify how much the automatic metrics (`same_root`, `is_similar_lev`) capture, on their own, of what the manual review (`synonym_manual`) judged to be a valid synonym among the non-exact answers.
- **Weighted Kappa (Landis & Koch 1977 scale)**, not raw kappa, to compare the three difficulty labels (declared by the author, cluster on accuracy, cluster on generation time). These are ordered categories; a hard/easy mistake is a worse disagreement than a hard/medium one.
- **Point-biserial correlation** (Classical Test Theory, borrowed from psychometrics) for the item discrimination index. For each item, the correlation between “the model solves it” and the model’s total score on the other 29 items.

- **1D k-means clustering** on accuracy per item, with silhouette score to establish how many groups are actually supported by the data, instead of assuming 3 or 5 a priori
- **Per-model bootstrap (2,000 resamples)** for the confidence interval of the size/accuracy correlation (RQ5).
- **Leave-one-out cross-validation** to choose the degree of fit for time/accuracy (RQ4/time as a proxy for difficulty). With N=30 items, a high-degree polynomial always improves in-sample R^2 without generalizing better. LOOCV measures true predictive power.
- **Shapiro-Wilk** (more suitable than Kolmogorov-Smirnov for small N) plus a Q-Q plot for the normality test of the accuracy distribution across models.

Reasoning Tokens

A separate notebook analyses the raw JSONL logs of the OpenRouter calls, which contain the native reasoning-token count provided by each provider. This makes it possible to empirically verify (not by inference from the model's name) which models actually reason, and to link generation duration to the number of tokens actually computed. Moreover, having the reasoning available lets us read how the model arrived at its answer.

Results and Comments

Presenting results and comments together calls for a different linguistic and communicative style. Presenting data in a plain, factual form and commenting on it in a discursive way makes it possible to analyse the numerical content and comment on it without academic formalism. In the previous part there was a greater degree of formalism in presenting information; in these sections the formalism is reduced in favour of greater clarity.

Descriptive Statistics of the Accuracy Distribution (models with verified parameters, n=45)

Basic descriptive statistics on the distribution of `exact_match` among the 45 models with a verified parameter count. The other models were excluded for not having a declared parameter count.

Statistic	Value
n models	45
Mean	0.2758
Median	0.2889
Standard deviation	0.1797
Variance	0.0323
Skewness	0.0707
Kurtosis (excess)	-1.0176
Minimum	0.0000
Maximum	0.6207
IQR (Q3-Q1)	0.2832

Table 3 Descriptive statistics of the models (N=45) with declared and/or known parameters

Skewness near zero (0.07) shows a distribution with no marked tail on either side. Mean and median are almost identical (0.276 vs. 0.289). Negative kurtosis (-1.02) indicates a flatter-than-normal distribution, with less concentration at the centre and more models spread toward the extremes than a normal distribution would predict. This is visually consistent with a list containing both near-zero models (*Hudson/gpt2*, *tinyllama*) and models under 60% (*gpt-oss:20b* at 57.8%, *qwen3.8:27b* at 54.4%), without a strong concentration in the middle of the ranking.

Normality Test

Shapiro-Wilk on the distribution gives $W = 0.9587$, $p = 0.1094$, compatible with normality ($p > 0.05$). Even so, the negative kurtosis seen above suggests not blindly trusting a single significance test on a sample of this size. Pairing it with a Q-Q plot for visual inspection shows where any deviations concentrate (tails or centre).

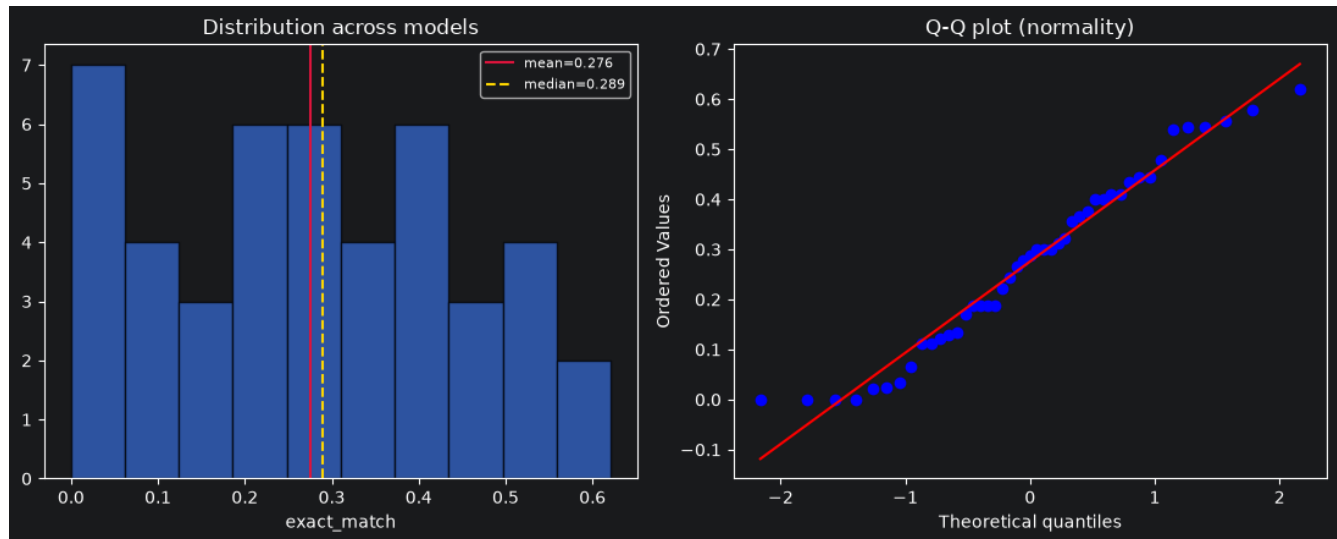


Figure 1 Distribution across models with declared parameters ($N=45$) on the left. Q-Q plot for normality on the right

The value of the preliminary analysis confirms that the shape is not pathologically distorted (bimodal, with extreme outliers), which would make it misleading to speak of an accuracy distribution across models. Most of the analysis uses non-parametric methods as an independent methodological choice (Spearman instead of Pearson, McNemar/Cochran's Q instead of normality-based tests). The preliminary analysis only confirms the choice of the analysis and methods used later on.

RQ1 / RQ2 — Does the author’s solution match the models’? How far apart are they, quantitatively?

Open-weight models (108, with 95% Wilson CI)

The model with the highest nominal `exact_match`, `z-ai/glm-5.2:free` (75.0%), must be discarded as “leader” since it has only 4 valid observations out of 90 expected (86 errors) and its 95% CI is notably wide (0.301–0.954). The table shows the ranking of the 10 models with the most exact answers (`exact_match`).

Model	n	exact_match	same_root	is_clue_echo	CI 95%
<i>stepfun/step-3.5-flash</i>	90	0.711	0.778	0.0	[0.610, 0.795]
<i>bytedance-seed/seed-2-1-turbo</i>	90	0.678	0.744	0.0	[0.576, 0.765]
<i>nvidia/nemotron-3-ultra-550b-a55b</i>	90	0.678	0.744	0.0	[0.576, 0.765]
<i>moonshotai/kimi-k2.5</i>	89	0.663	0.73	0.0	[0.56, 0.753]
<i>xiaomi/mimo-v2.5-pro</i>	90	0.656	0.722	0.0	[0.553, 0.746]
<i>qwen/qwen3.7-plus</i>	89	0.629	0.697	0.0	[0.525, 0.722]
<i>olmo-3.1:32b</i>	58	0.621	0.69	0.0	[0.492, 0.734]
<i>z-ai/glm-4.7</i>	88	0.614	0.659	0.0	[0.509, 0.709]
<i>deepseek/deepseek-r1-0528</i>	90	0.611	0.678	0.0	[0.508, 0.705]
<i>tencent/hy3</i>	90	0.6	0.667	0.0	[0.497, 0.695]

Table 4 Top 10 open-weight models ranked by exact answers (`exact_match`)

At the bottom of the ranking, five models at `exact_match` = 0.000: `tinylama:1.1b`, `starcoder2:15b`, `granite4:350m`, `Hudson/gpt2-instruct:345m-q8_0`, `meta-llama/llama-3.2-1b-instruct`.

Three of these five have a very high `is_clue_echo` rate, confirming that the task was not solved, but rather that they repeat one of the given clues instead of finding the connector. `Starcoder2:15b`, despite failing, has a low `is_clue_echo` (0.022) but `format_compliant` = 0.000, consistent with the injection of unrelated text/code. `Meta-llama/llama-3.2-1b-instruct` has `format_compliant` = 0.978 (it respects the required format) but `is_clue_echo` only 0.189 — this shows that its failure is neither echo nor format violation, it simply does not find the right word.

Model	n	exact_match	same_root	is_clue_echo	CI 95%
<i>tinylama:1.1b</i>	90	0.0	0.0	0.767	[0.0, 0.041]
<i>starcoder2:15b</i>	89	0.0	0.0	0.022	[0.0, 0.041]
<i>granite4:350m</i>	90	0.0	0.0	0.611	[0.0, 0.041]
<i>Hudson/gpt2-instruct:345m-q8_0</i>	90	0.0	0.022	0.7	[0.0, 0.041]
<i>meta-llama/llama-3.2-1b-instruct</i>	90	0.0	0.0	0.189	[0.0, 0.041]
<i>stablelm2:1.6b-chat</i>	90	0.022	0.022	0.744	[0.006, 0.077]
<i>deepseek-r1:1.5b</i>	85	0.024	0.047	0.447	[0.006, 0.082]
<i>orca-mini:3b</i>	90	0.033	0.044	0.778	[0.0110, 0.093]
<i>minicpm-v4.6:1b</i>	90	0.067	0.1	0.489	[0.0310, 0.138]
<i>ministral-3:3b</i>	90	0.111	0.156	0.0	[0.0610, 0.193]

Table 5 Bottom 10 open-weight models ranked by exact answers (exact_match)

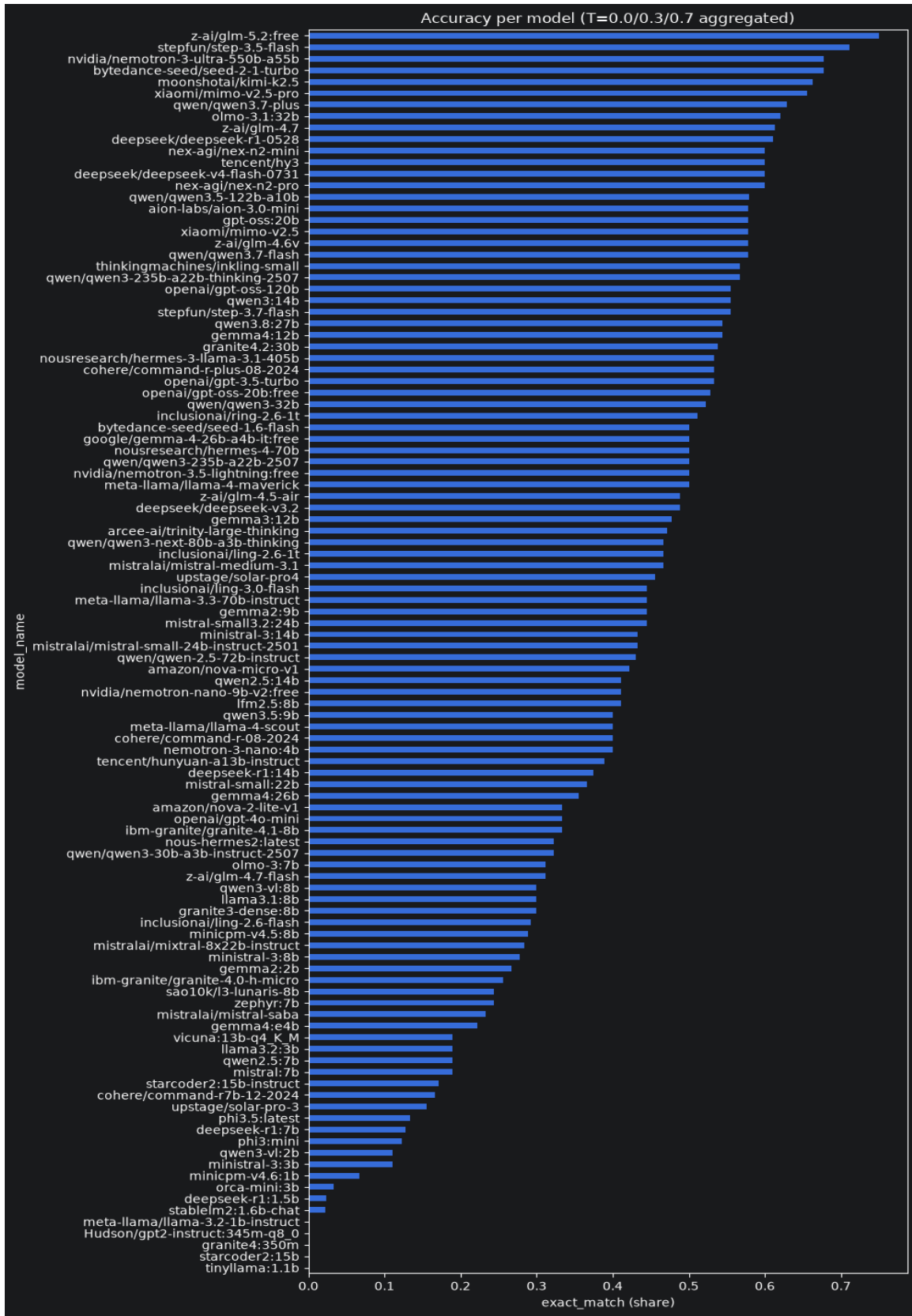


Figure 2 Exact_match for open-weight models with data aggregated by temperature

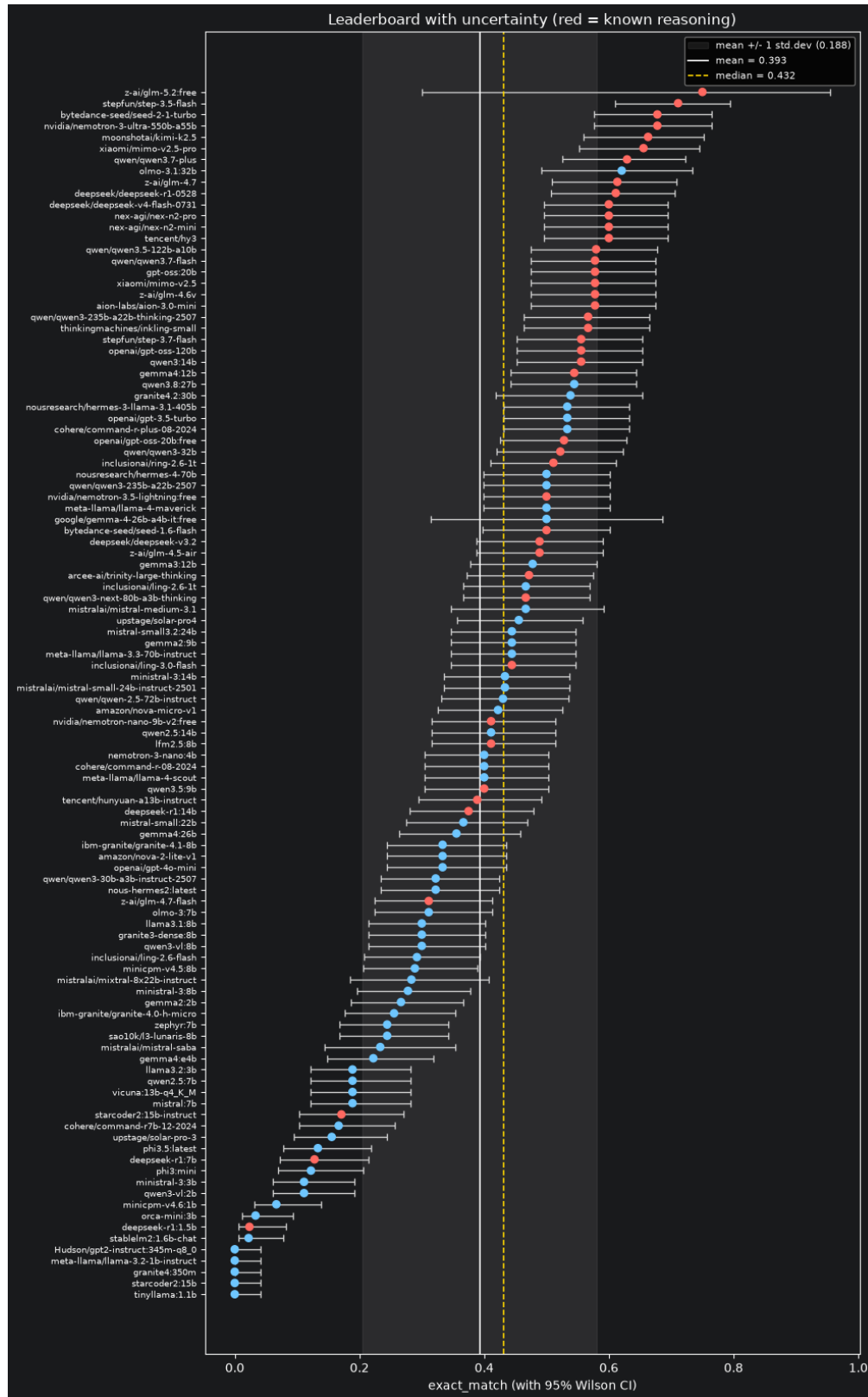


Figure 3 Ranking with uncertainty levels and confidence index (CI95%). Dashed yellow line = median. Solid white line = mean. The shaded grey area covers one standard deviation. Models with a red dot indicate reasoning models. Horizontal bars = confidence index

Commercial models (16, 95% Wilson CI, N=30 per model)

Model	n	exact_match	same_root	is_clue_echo	CI 95%
<i>x-ai/grok-4.6</i>	30	0.767	0.833	0.0	[0.591, 0.882]
<i>google/gemini-3.7-flash</i>	30	0.733	0.800	0.0	[0.556, 0.858]
<i>anthropic/claude-opus-5</i>	30	0.700	0.767	0.0	[0.521, 0.833]
<i>moonshotai/kimi-k3</i>	30	0.700	0.767	0.0	[0.521, 0.833]
<i>openai/gpt-5.6-sol</i>	30	0.700	0.767	0.0	[0.521, 0.833]
<i>openai/gpt-5.5</i>	30	0.667	0.733	0.0	[0.488, 0.808]
<i>anthropic/claude-sonnet-5</i>	30	0.667	0.733	0.0	[0.488, 0.808]
<i>z-ai/glm-5.3</i>	30	0.667	0.733	0.0	[0.488, 0.808]
<i>deepseek/deepseek-v4-pro-0813</i>	30	0.633	0.700	0.0	[0.455, 0.781]
<i>inception/mercury-2</i>	24	0.583	0.667	0.0	[0.388, 0.755]
<i>anthropic/claude-haiku-4.5</i>	30	0.567	0.633	0.0	[0.392, 0.726]
<i>qwen/qwen3.8-max</i>	30	0.567	0.633	0.0	[0.392, 0.726]
<i>cohere/command-a</i>	30	0.533	0.600	0.0	[0.361, 0.698]
<i>openai/gpt-5.4</i>	30	0.533	0.600	0.0	[0.361, 0.698]
<i>openai/gpt-5.4-mini</i>	30	0.433	0.500	0.0	[0.274, 0.608]

Table 6 Ranking of all commercial models by exact_match

meta/muse-spark-1.2 failed all 30 calls (excluded from the ranking).

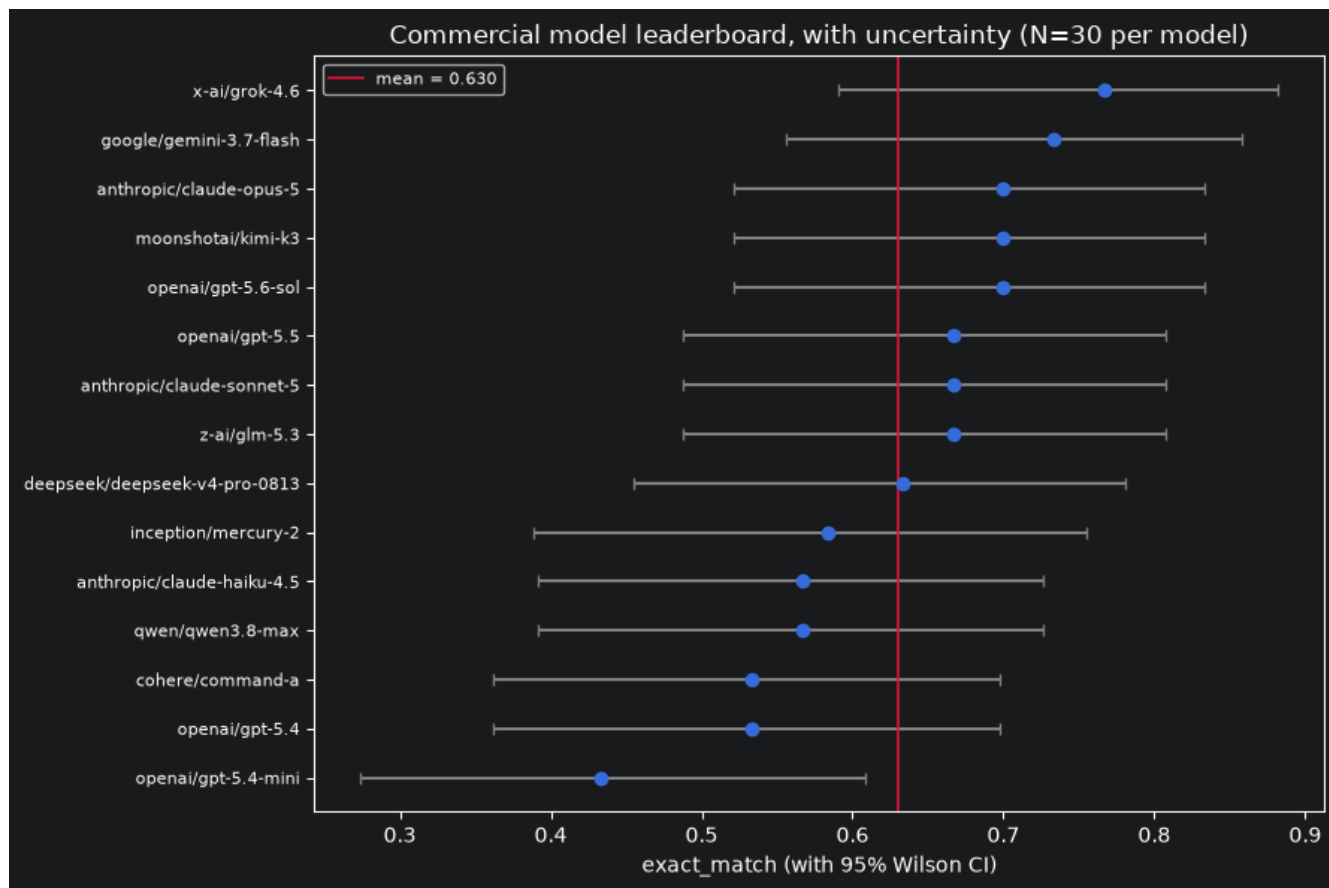


Figure 4 Ranking with uncertainty levels and confidence index (CI95%). Solid red line = mean.
Horizontal bars = confidence index

Extended Correctness (exact + same_root + manual synonym)

Including hand-validated synonyms as well, the picture changes slightly: *bytedance-seed/seed-2-1-turbo* rises to first place (80.0% extended correctness, *broad_correct*), followed by *nvidia/nemotron-3-ultra-550b-a55b* (78.9%) and *stepfun/step-3.5-flash* (78.9%).

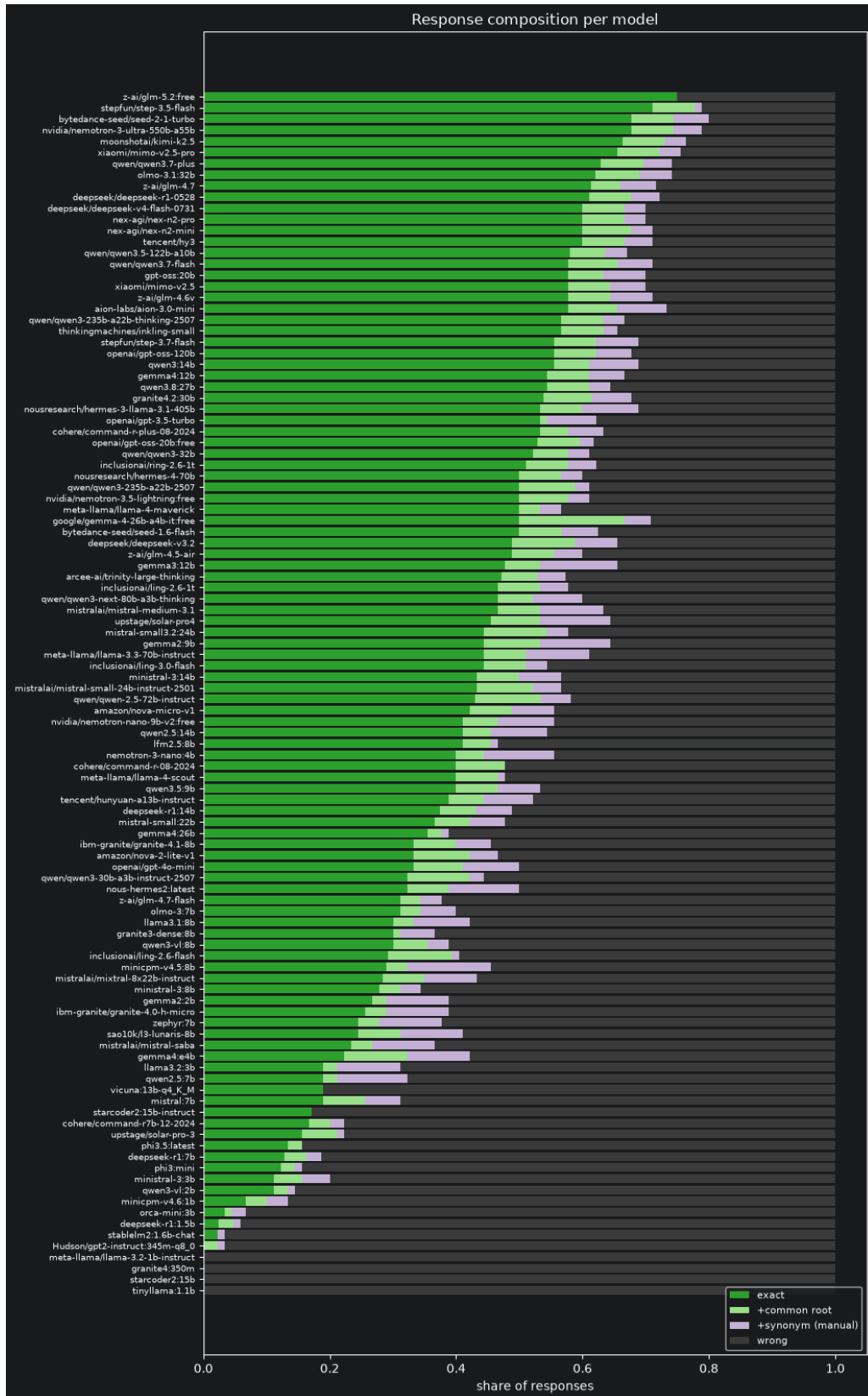


Figure 5 Composition of answers for open-weight models with aggregated data. Dark green = *exact_match*. Light green = *common root*. Pink = *synonym*. Grey = *wrong*.

Version for commercial models only

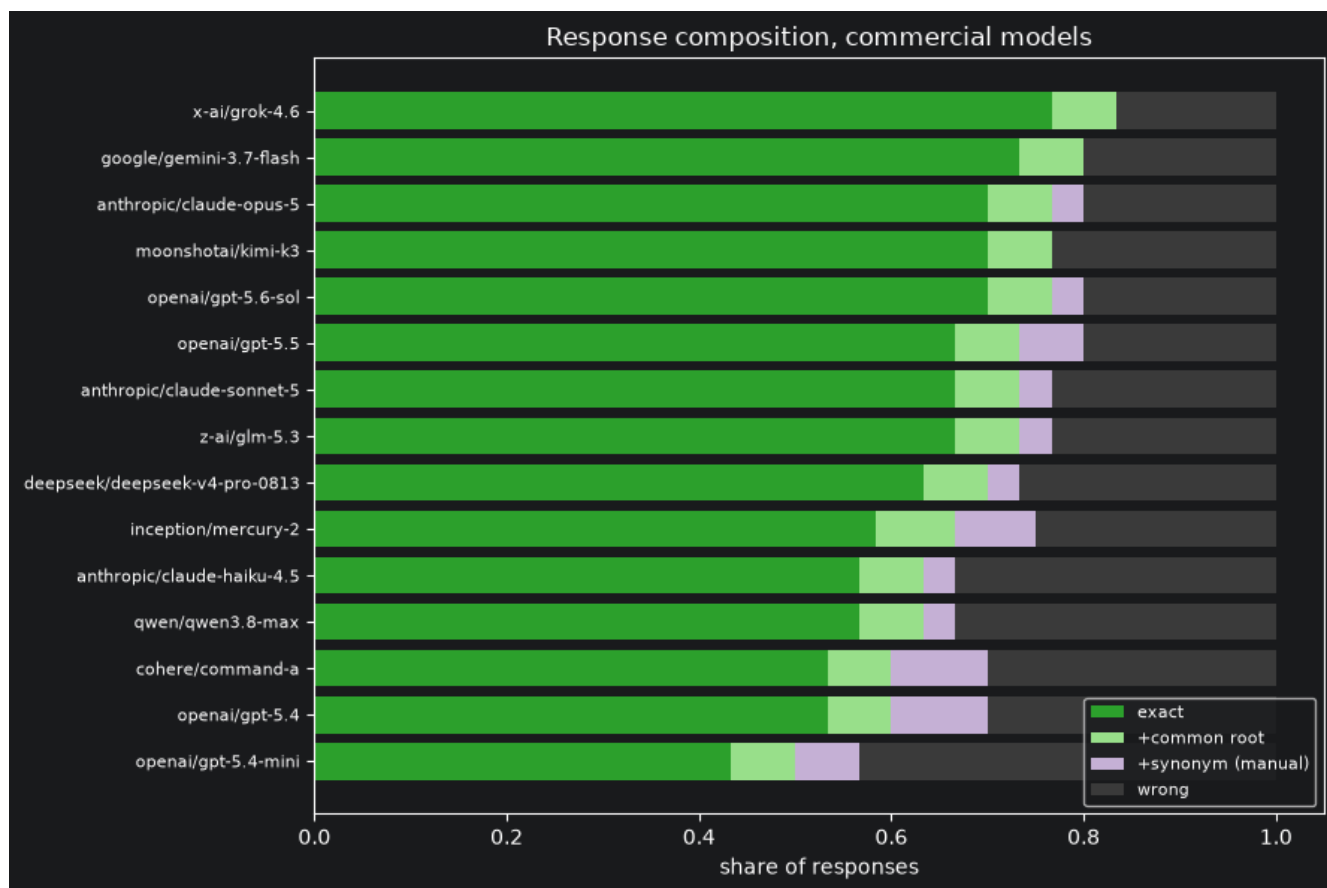


Figure 6 Composition of answers for commercial models with aggregated data. Dark green = *exact_match*. Light green = *common root*. Pink = *synonym*. Grey = *wrong*.

The Importance of Manual Review

The agreement between the automatic metrics (*same_root* or *is_similar_lev*) and the manual synonym judgement, among non-exact answers, gives a Cohen's Kappa = -0.041 (raw agreement 81.9%; kappa corrects for agreement due to chance). The automatic metrics alone would not have captured 468 valid synonyms out of 5,732 non-exact rows reviewed. The full manual review confirms its real usefulness and statistical value, not a redundant precautionary check.

Pairwise Comparison Among the Top Models (McNemar + FDR)

Among the top 8 open-weight models, 0 pairs out of 21 remain significantly different after FDR correction ($p < 0.05$). The same outcome holds among the top 8 commercial models, with 0 pairs out of 28. In both groups, with $N \text{ items} = 30$, the nominal best model is not statistically distinguishable from the other top-ranked models — a point relevant to avoid over-interpreting the exact ranking position of the leading models.

RQ3 — Can a model’s answer turn out to be more coherent than the author’s?

Mathematically, it is difficult to rigorously demonstrate whether LLM answers can be more valid or coherent than human ones; this kind of answer is a qualitative rather than quantitative one.

Both manual evaluation columns are filled in 100% (9,380/9,380 open-weight rows, 444/444 commercial). The models with the highest rate of answers judged a valid synonym (synonym_manual) among the open-weight ones are *gemma2:9b* (14.4%), *minicpm-v4.5:8b* (13.3%), *gemma3:12b* (13.3%). For linguistic coherence (coherence_manual) the highest rate is *gemma3:12b* (23.3%), followed by *ibm-granite/granite-4.0-h-micro* (21.1%) and *minicpm-v4.5:8b* (21.1%).

Qualitatively, the candidates worth focusing on are:

- Guillo_21 (supermarket with checkout). The most common answer is not checkout (30.0%) but store (43.2%); the latter uses the author’s strategy “everything is inside a shop/supermarket.” It is only a more generic word, a vaguer synonym, not a better idea. Checkout, on the other hand, is conceptually different. It is not a container, but a process interpretable as a phase of the process (exit, you leave through the checkout; shelves lead to the checkout; cart, used up until the checkout; pay is the checkout; customer is whoever goes through the checkout). We can identify checkout as the better candidate, not by percentage of responses given, but by association with the behaviours or tools used. Both can be considered valid to the same degree.
- Guillo_22 (Emotion with stress): emotion is given by only 3.9% of attempts, while stress dominates at 33.8%. Several reasoning chains arrive at STRESS through systematic elimination of alternatives (reflex, stimulus, impulse...). The most convincing pattern is provided by *gpt-oss-120b*, which states: “Reaction: ‘stress reaction’ yes. Response: ‘stress response’. Driven: ‘stress-driven’. Cognitive: ‘cognitive stress’. Adrenaline: ‘adrenaline stress’? Usually...” Four very strong idiomatic collocations (stress reaction, stress response, stress-driven, cognitive stress), with only the last one slightly forced, as flagged by the model itself. The original justification in the dataset is objectively weaker and somewhat circular (“emotion reach a reaction and a response, emotion drive our behaviour...”, using “emotion” to explain why the answer is “emotion”). Stress is a tighter association compared to the author’s.
- Guillo_18 (watching with watch). Here it is not a disagreement over the concept, but a disagreement over the form in which the word is written. Practically every model answers watch, not watching (198/339, 58.4% in the open-weight set alone, same_root=1). The association (game watching, TV watching, bird watching, vigilante watching) is correct 100% of the time. In hindsight, the gerund is the least natural form for the connector word, and almost no model lands on it. The item confirms the correctness of the association, but the choice of the solution’s grammatical form made the item nearly impossible to hit exactly.
- In these cases, the majority of models produce a tighter clue-by-clue explanation. After manual review, in both cases the original justification in the dataset is among the weakest or most generic.

Cases dismissed:

- Guillo_20 (travel vs vacation, 36% vs 27%): vacation is nearly a synonym of travel given these clues (exotic, adventure, fly, culture, relax); it is not a different or better idea, it is the same idea with another word.
- Guillo_24 (idea vs light/solution): light wins only on the clue “lamp” (via the idiom lightbulb moment) but does not hold up on construct/problem/innovation/good the way idea does. Idea remains the more solid answer.
- Guillo_23 (concept): zero models out of 487 attempts ever answered “concept,” not even the best commercial ones. The dominant alternatives (theory 25.9%, science 25.7%) are not clearly better: “business theory” is less natural than “business concept,” “society theory” is unusual. This is an anomalous case in that it is not about finding a more suitable word — the category is abstract and no single word truly wins — it is an item-design problem rather than a failure of the models’ ability to extract the right word.
- guillo_29 (scraping): spider (18.9%) captures only the clue “web,” not the other four. This is an example of anchoring on an association using only part of the given clues, not a complete alternative solution.

RQ4 — How much does a model’s answer change across the temperature spectrum?

Evaluating the answers across the temperature spectrum, we obtain an accuracy of 39.3% at $T=0.0$ (1,201/3,057), 38.7% at $T=0.3$ (1,219/3,152), 38.7% at $T=0.7$ (1,228/3,171). This difference is not statistically significant: aggregate Cochran’s $Q = 0.558$, $p = 0.757$ (3,023 model $\times$ item pairs with all 3 temperatures, 434 with a variable outcome). These data confirm that temperature is not a discriminating parameter for assessing the correctness of answers.

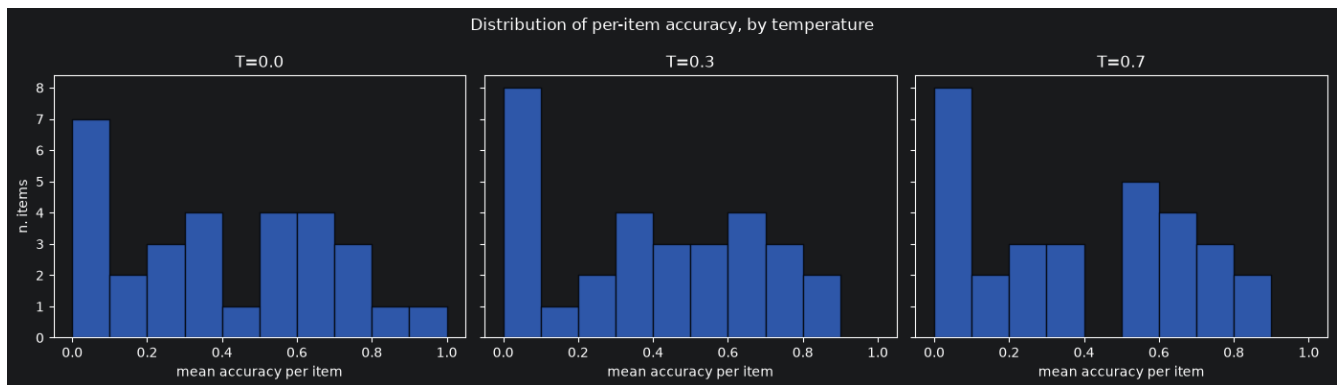


Figure 7 Distribution of accuracy per item across the temperature spectrum ($T=0.0/0.3/0.7$)

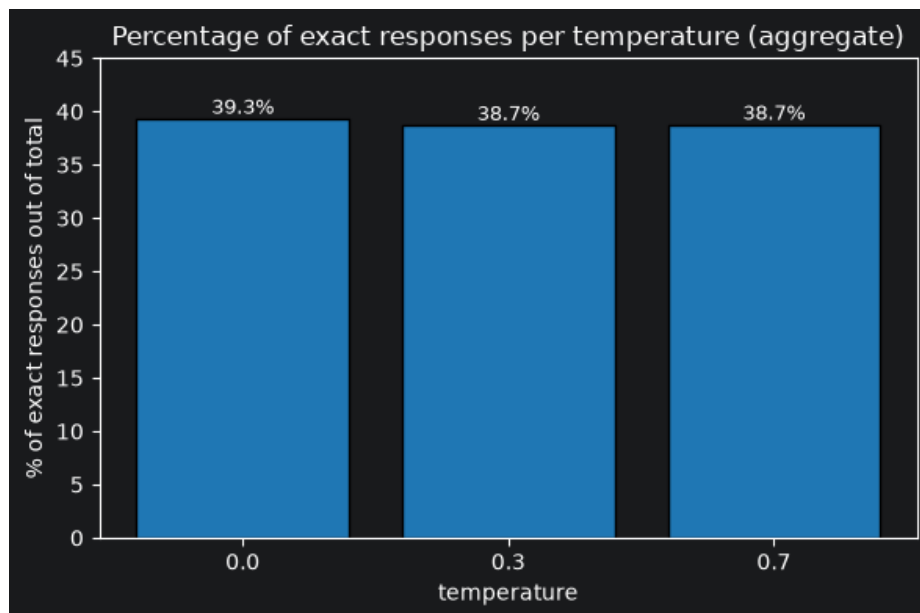


Figure 8 Aggregated data on the percentage of exact answers across the temperature spectrum.

Per-Model Test Across the Temperature Spectrum

12 models are not testable, since no item changes outcome between temperatures for them, meaning there is no variance to test. Among the testable models, 9 turn out sensitive to temperature with $p < 0.05$, without correction for multiple comparisons: *gemma4:26b* ($Q=12.20$, $p=0.0022$), *gemma4:12b* ($Q=7.60$, $p=0.0224$), *nousresearch/hermes-4-70b* ($Q=7.00$, $p=0.0302$), *gemma4:e4b* ($Q=6.89$, $p=0.0319$), *deepseek-r1:7b* ($Q=6.33$, $p=0.0421$), *tencent/hunyuan-a13b-instruct* ($Q=6.22$, $p=0.0446$), *starcode2:15b-instruct*, *ministral-3:14b* and *thinkingmachines/inkling-small* ($Q=6.00$, $p=0.0498$ for all three).

By magnitude of shift (max–min accuracy across the three T’s): *z-ai/glm-5.2:free* and *google/gemma-4-26b-a4b-it:free* (0.333 for both), *gemma4:26b* (0.300), *tencent/hunyuan-a13b-instruct* and *gemma4:e4b* (0.200 for both).

The models’ accuracy ranking stays stable across temperatures. Spearman correlations between rankings: $T=0.0/T=0.3 = 0.920$, $T=0.0/T=0.7 = 0.886$, $T=0.3/T=0.7 = 0.930$ (all with $p < 0.0001$).

There are models more sensitive to temperature, shifting absolute accuracy by a few points. This is a signal of reliable measurement and response flexibility. The calculation does not apply to commercial models, which have a single run and no temperature axis.

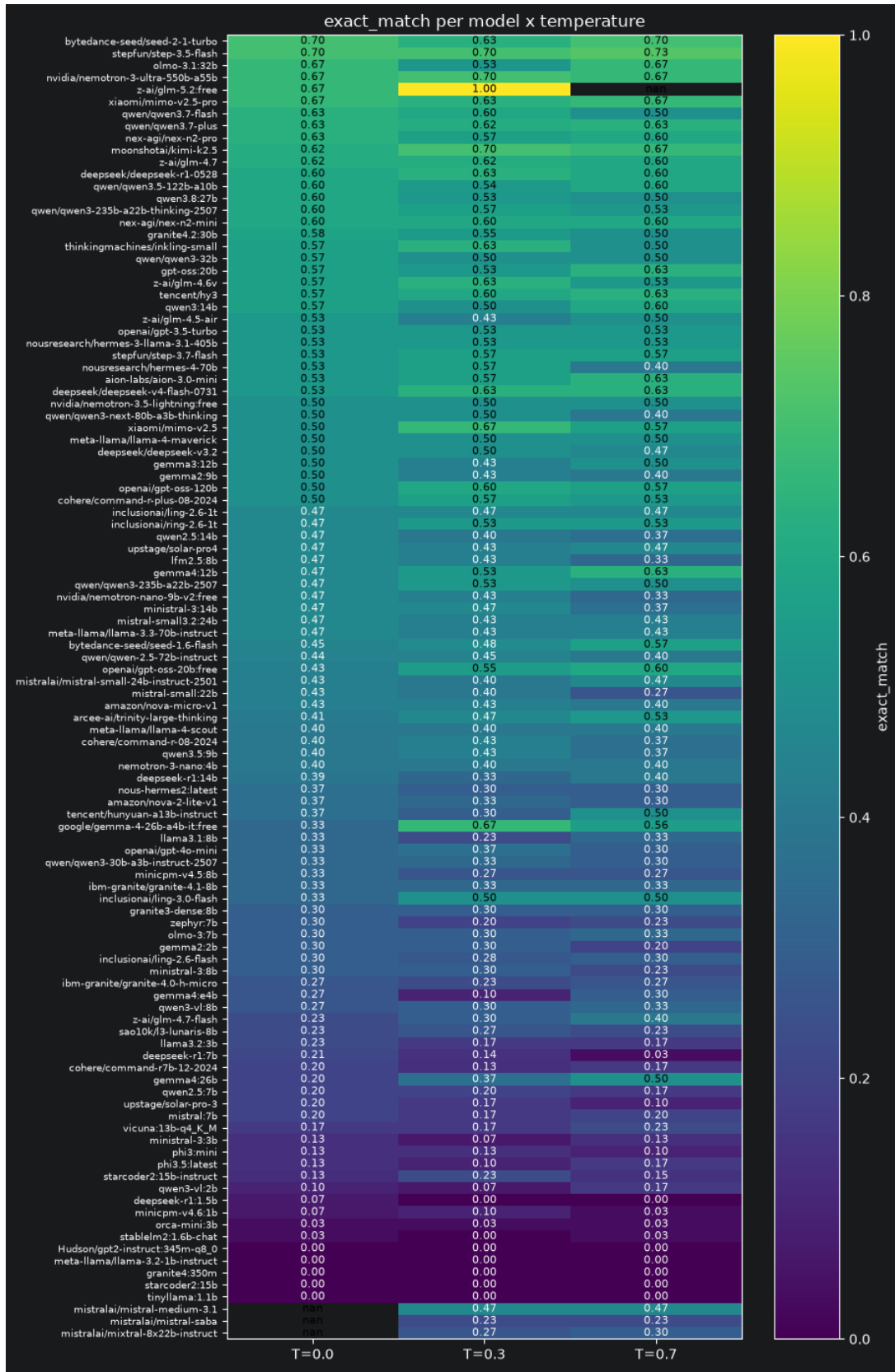


Figure 9 Heatmap using exact_match for open-weight models across the temperature spectrum.

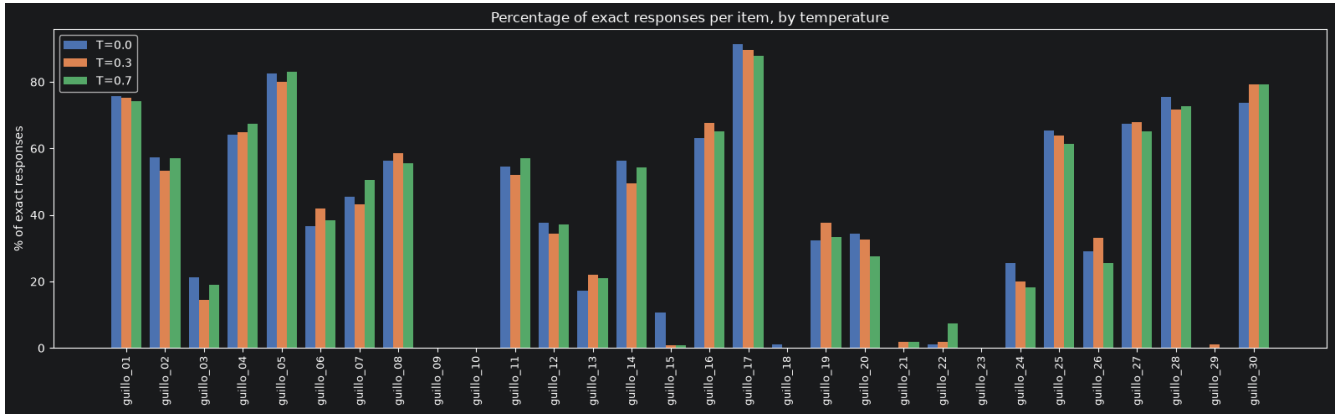


Figure 10 Percentage of exact answers (*exact_match*) ordered by item. Blue = $T=0.0$, Orange = $T=0.3$, Green = $T=0.7$.

Overall, one observes a similarity of exact answers across the three set temperatures for each item, confirming that the general trend of the answers is not determined by temperature.

RQ5 — Does model size correlate with the accuracy of the association?

The number of parameters represents a model’s potential capacity for abstraction and elaboration. Given equal training-data quality, a larger number of parameters allows a model to grasp more complex nuances, connect different subjects, and develop deeper reasoning rather than stopping at basic concepts.

Some models report their effective size in their tag, based on parameter count. 41 were excluded because the tag does not report a size. Through systematic research, the declared parameters of other models were found, though not for all the ones used. Based on the available data, we can identify a total of 70 models with declared parameters. Applying the relationship between parameters and `exact_match`, we obtain a Spearman $\rho = 0.827$ (all models), $\rho = 0.842$ (non-reasoning only). Bootstrap 95% CI (2,000 resamples) [0.701, 0.910], indicating a strong and reasonably precise correlation.

Deepening the analysis with parameter counts verified against official model cards/technical reports (45 models, not derived from the tag), we obtain $\rho = 0.756$ (total checkpoint parameters) versus $\rho = 0.690$ for active parameters per token — relevant for Mixture-of-Experts models or ones with Per-Layer Embeddings, where the two numbers diverge considerably. The clearest case, *gpt-oss:20b*, has 20.9B total parameters but only 3.61B active per token (MoE, 32 experts, top-4 routing), yet `exact_match` = 0.578. The size/accuracy correlation is therefore solid but does not explain everything. Just over half of the models with declared parameters is not enough to confirm the exactness of this calculation or to fully answer the question. The actual per-token compute matters, not just the declared checkpoint size.

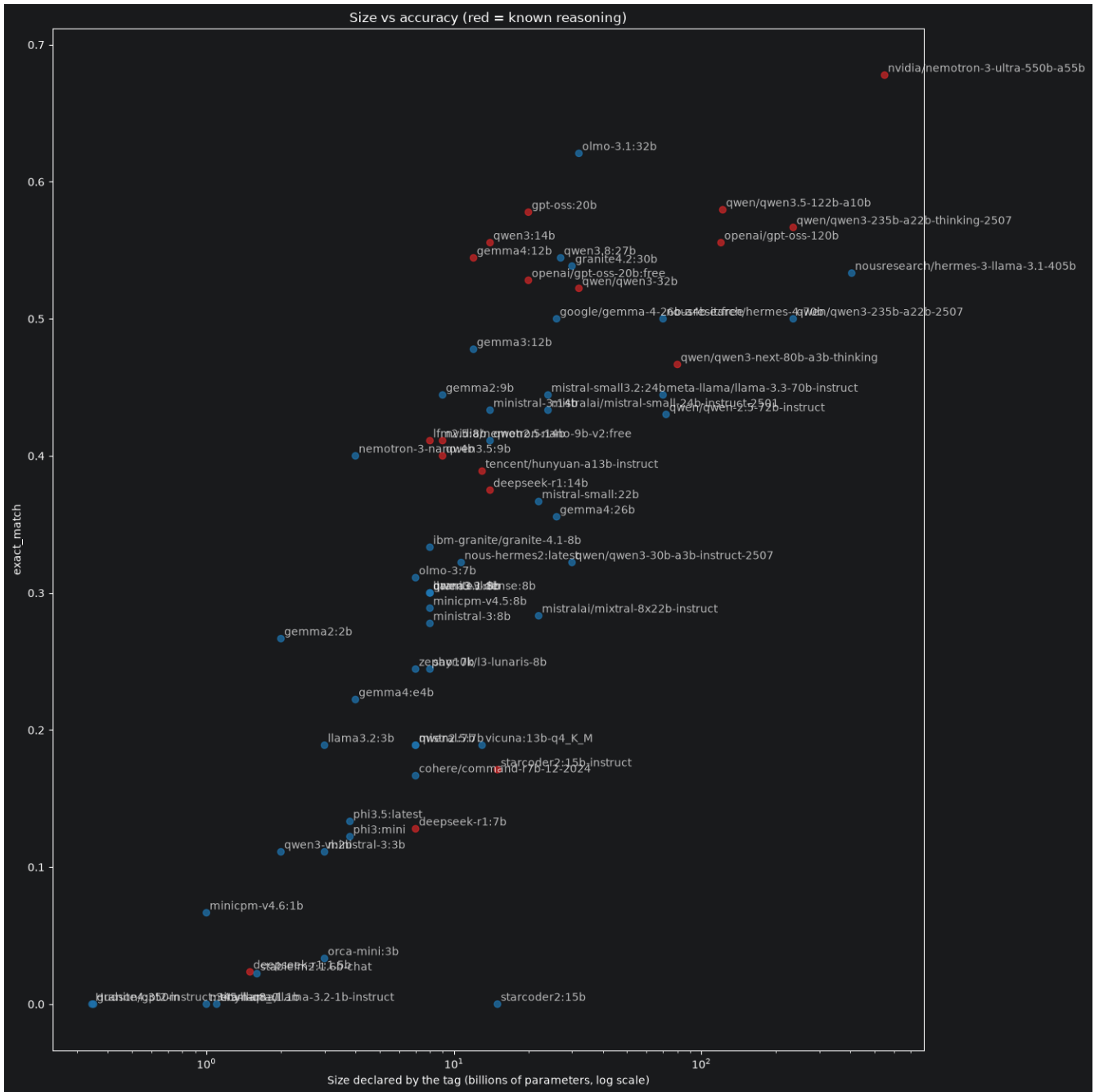


Figure 11 Size vs. accuracy of models with declared or estimated parameters. In red, models with known reasoning

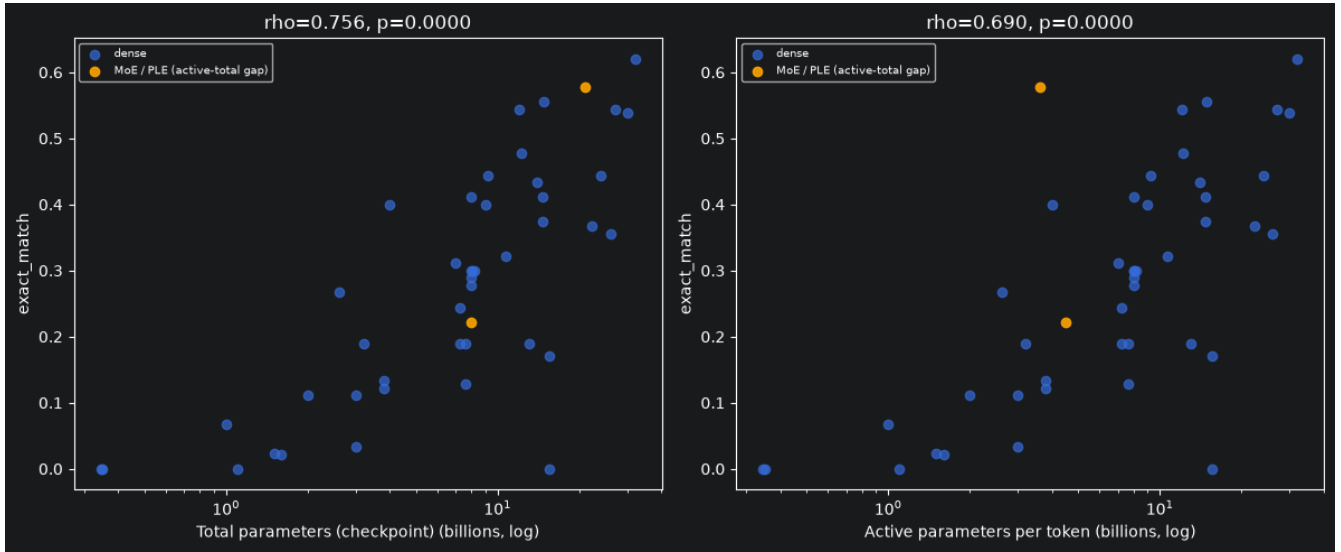


Figure 12 Total parameters (checkpoint) vs. active parameters per token ($\rho=0.756$ and $\rho=0.690$), in yellow the MoE models

RQ6 — Which model gives the best answer, according to the author’s judgement?

This question, by its very construction, is a qualitative judgement that cannot be derived from aggregated numbers. The final data provide the objective basis on which to ground the choice. Further analyses are necessary to be able to provide an objectively solid answer.

Calibration, Discrimination, Empirical Classification of Items

The hardest and easiest items were identified from the open-weight models’ processing by calculating how often they gave an exact answer. Average accuracy across all 108 models identified the following as the hardest items:

- guillo_10 (pc, 0.000)
- guillo_09 (petrichor, 0.000)
- guillo_23 (concept, 0.000)
- guillo_18 (watching, 0.003)
- guillo_29 (scraping, 0.003)
- guillo_21 (supermarket, 0.013)
- guillo_22 (emotion, 0.035)
- guillo_15 (statistic, 0.041)
- While the easiest items:
- guillo_17 (flag, 0.895)
- guillo_05 (light, 0.818)
- guillo_30 (present, 0.775)

The same pattern is found, almost identically, among the commercial models. The hardest items are near or practically at 0.000:

- guillo_15
- guillo_10
- guillo_22
- guillo_18
- guillo_29
- guillo_23

The Spearman correlation between per-item difficulty as seen by the 108 open-weight models and as seen by the 16 commercial ones is $\rho = 0.866$ ($p < 0.0001$). The notion of a “hard item” is the same regardless of model tier — a relevant convergent-validity result, independent of model type, size, or reasoning capability.

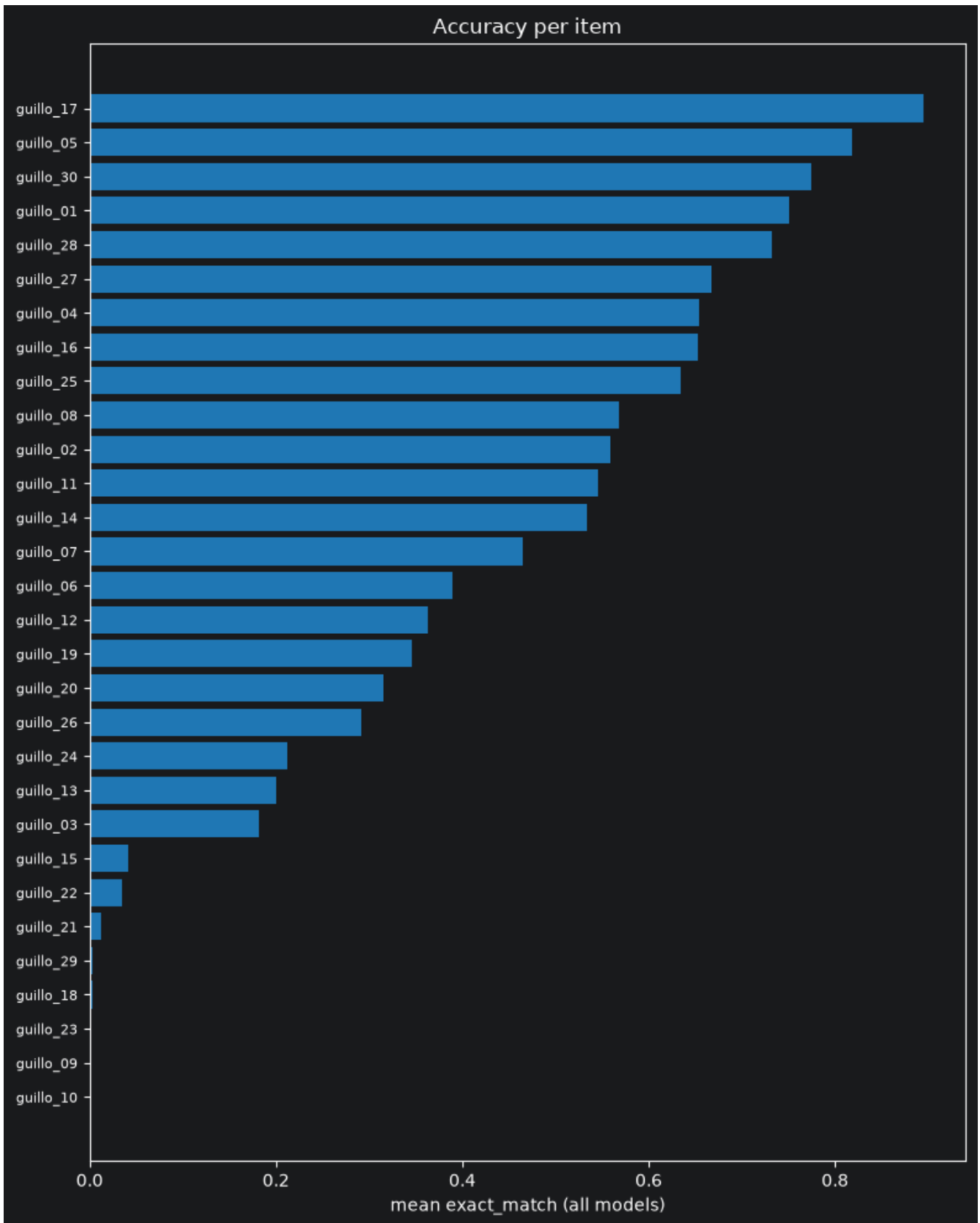


Figure 13 Accuracy per item with aggregated data including both open-weight and commercial models

The calibration of the difficulty labels declared a priori by the author (`difficulty_note`) shows an average accuracy of 0.487 for “easy” (1,250 rows), 0.407 for “medium” (5,010 rows), 0.320 for “hard” (3,120 rows); the ordering is as expected, but with wide overlap between bands.

Wanting to expand the number of classes beyond the three provided, the silhouette score does not indicate a clearly optimal k : $k=2$: 0.650, $k=3$: 0.669, $k=4$: 0.656, $k=5$: 0.670, $k=6$: 0.701. There is an increase with k but no marked elbow. At the same granularity as the 3 labels ($k=3$), the empirical boundaries fall at `exact_match` = 0.111 (hard–medium) and 0.499 (medium–easy). 18 items out of 30 (60%) disagree with the author’s label and the empirical cluster, with a substantial divergence.

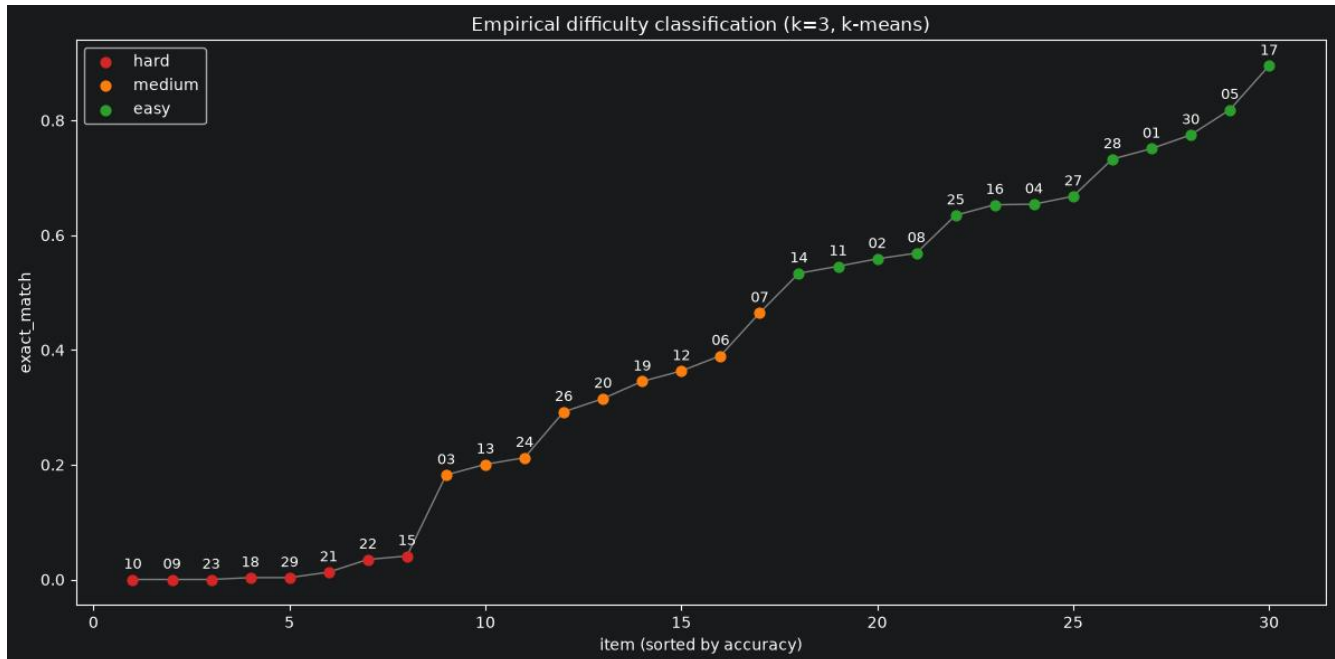


Figure 14 Classification with $k=3$, k -means, based on the hard, medium, easy labels given by the author

Through visual inspection of the chart, the author hypothesised a $k=5$ classification. The hypothesised boundaries: very hard–hard = 0.111; hard–medium = 0.252; medium–easy = 0.499, easy–very easy = 0.700. This further subdivision of difficulty aims only to identify a different split, identifying two additional extremes (very hard and very easy).

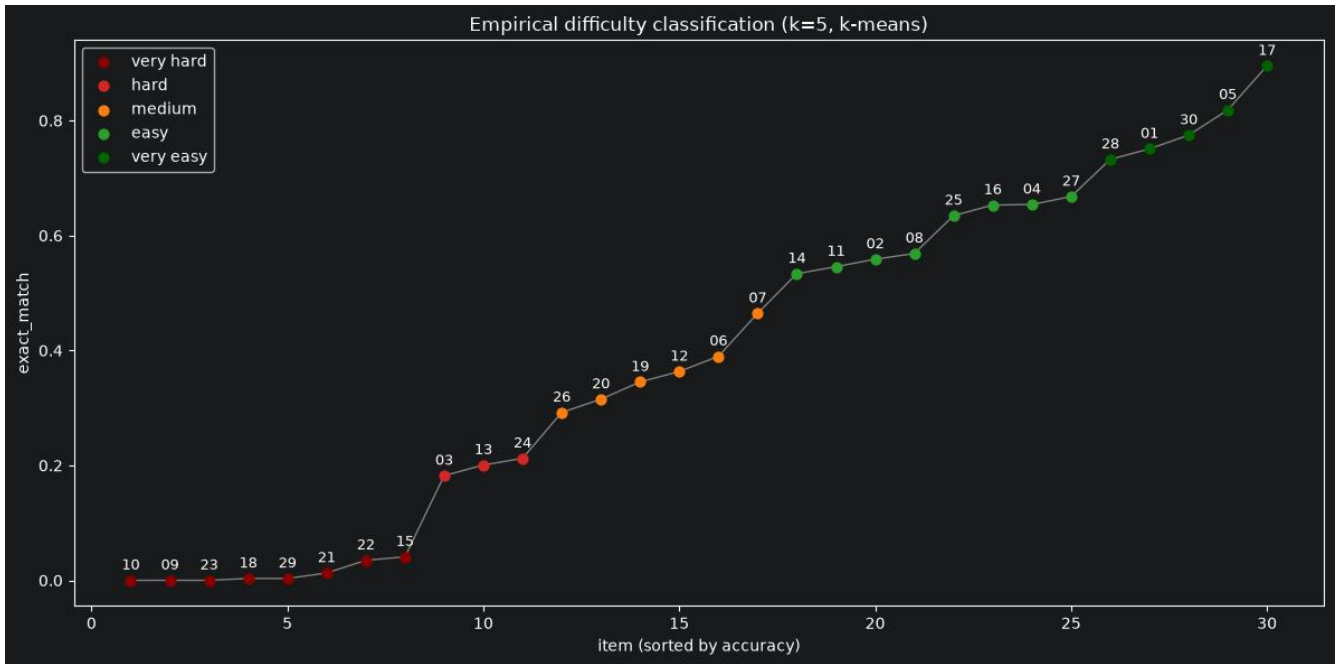


Figure 15 Classification with $k=5$, k -means, based on the very hard, hard, medium, easy, very easy labels hypothesised by the author

Looking at the chart, further subdivisions could be made, but as mentioned earlier there would be no marked differences between the categories.

The discrimination index using point-biserial correlation from Classical Test Theory identified 3 items not testable due to lack of variance: guillo_09, guillo_10, guillo_23. Lower or negative discrimination, flagging candidates for item review rather than model review: guillo_15 (-0.161), guillo_18 (-0.146), guillo_22 (-0.094), guillo_03 (0.069), guillo_21 (0.122).

The convergent validity of the three difficulty measures (declared label, accuracy cluster, generation-time cluster of the reasoning models only) gives a very low weighted kappa:

- label-vs-accuracy = 0.122 (slight)
- label-vs-time = 0.145 (slight)
- accuracy-vs-time = 0.325 (fair)

Generation time agrees reasonably well with accuracy, identifying a difficulty genuinely experienced by the model, but agrees almost not at all with the author's label — related but non-overlapping constructs. This is not an error to be corrected but an interesting result for understanding how human and AI-model interpretation differ.

Item	difficulty_note	cluster_accuracy	cluster_time	exact_match	duration_mean (sec)	is_clue_echo
guillo_01	easy	easy	easy	0.751	47.44	0.064
guillo_02	easy	easy	easy	0.559	60.571	0.07
guillo_03	medium	medium	hard	0.182	229.136	0.064
guillo_04	medium	easy	easy	0.654	57.284	0.105
guillo_05	medium	easy	easy	0.818	23.961	0.041
guillo_06	medium	medium	easy	0.39	53.384	0.048
guillo_07	medium	medium	easy	0.465	84.72	0.116
guillo_08	hard	easy	easy	0.569	56.83	0.08
guillo_09	hard	hard	hard	0.0	295.052	0.084
guillo_10	easy	hard	easy	0.0	62.85	0.058
guillo_11	medium	easy	medium	0.546	162.469	0.065
guillo_12	medium	medium	easy	0.363	99.316	0.077
guillo_13	hard	medium	easy	0.201	70.938	0.111
guillo_14	hard	easy	easy	0.534	36.83	0.083
guillo_15	medium	hard	easy	0.041	9.231	0.06
guillo_16	hard	easy	easy	0.653	25.465	0.079
guillo_17	medium	easy	easy	0.895	7.768	0.057
guillo_18	medium	hard	easy	0.003	64.501	0.058
guillo_19	hard	medium	easy	0.345	70.994	0.074
guillo_20	medium	medium	medium	0.315	156.197	0.048
guillo_21	medium	hard	easy	0.013	82.955	0.08
guillo_22	medium	hard	medium	0.035	146.481	0.177
guillo_23	hard	hard	medium	0.0	186.665	0.029
guillo_24	hard	medium	medium	0.212	191.056	0.061
guillo_25	easy	easy	easy	0.635	77.862	0.096
guillo_26	medium	medium	medium	0.292	127.042	0.092
guillo_27	hard	easy	easy	0.668	57.75	0.022
guillo_28	medium	easy	easy	0.732	28.712	0.07
guillo_29	hard	hard	hard	0.003	260.927	0.156
guillo_30	medium	easy	easy	0.775	27.909	0.102

Table 7 Comparison between the human classification and the models' classification based on task-execution time. Highlighted in bold the item with the longest and the shortest processing duration

Below, a graphical visualization of the preceding table.

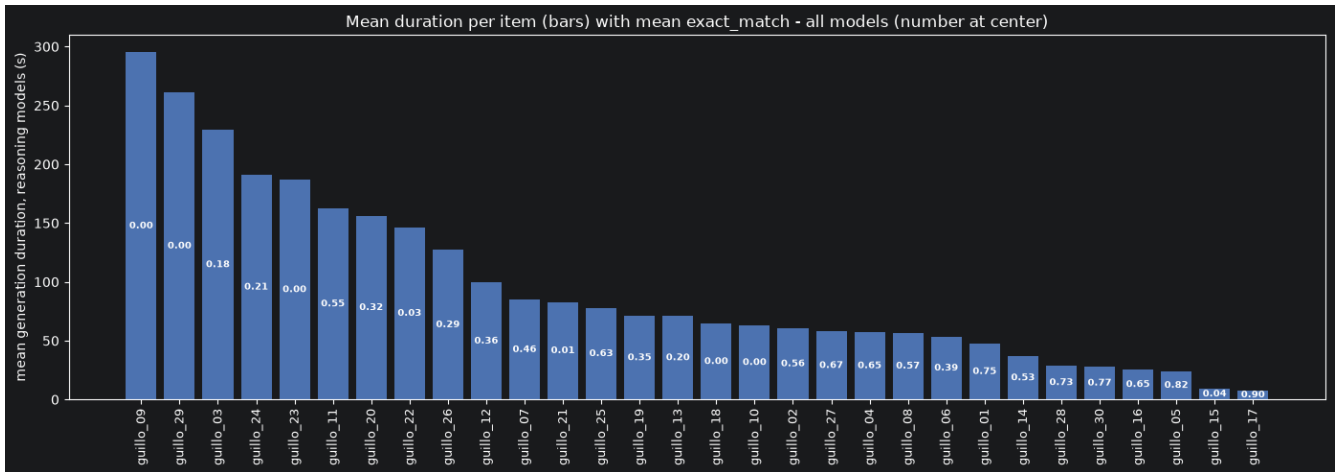


Figure 16 Average duration per item (bars) in seconds, with average exact match at the centre (ordered by duration)

The heat map contains the Spearman rho data.

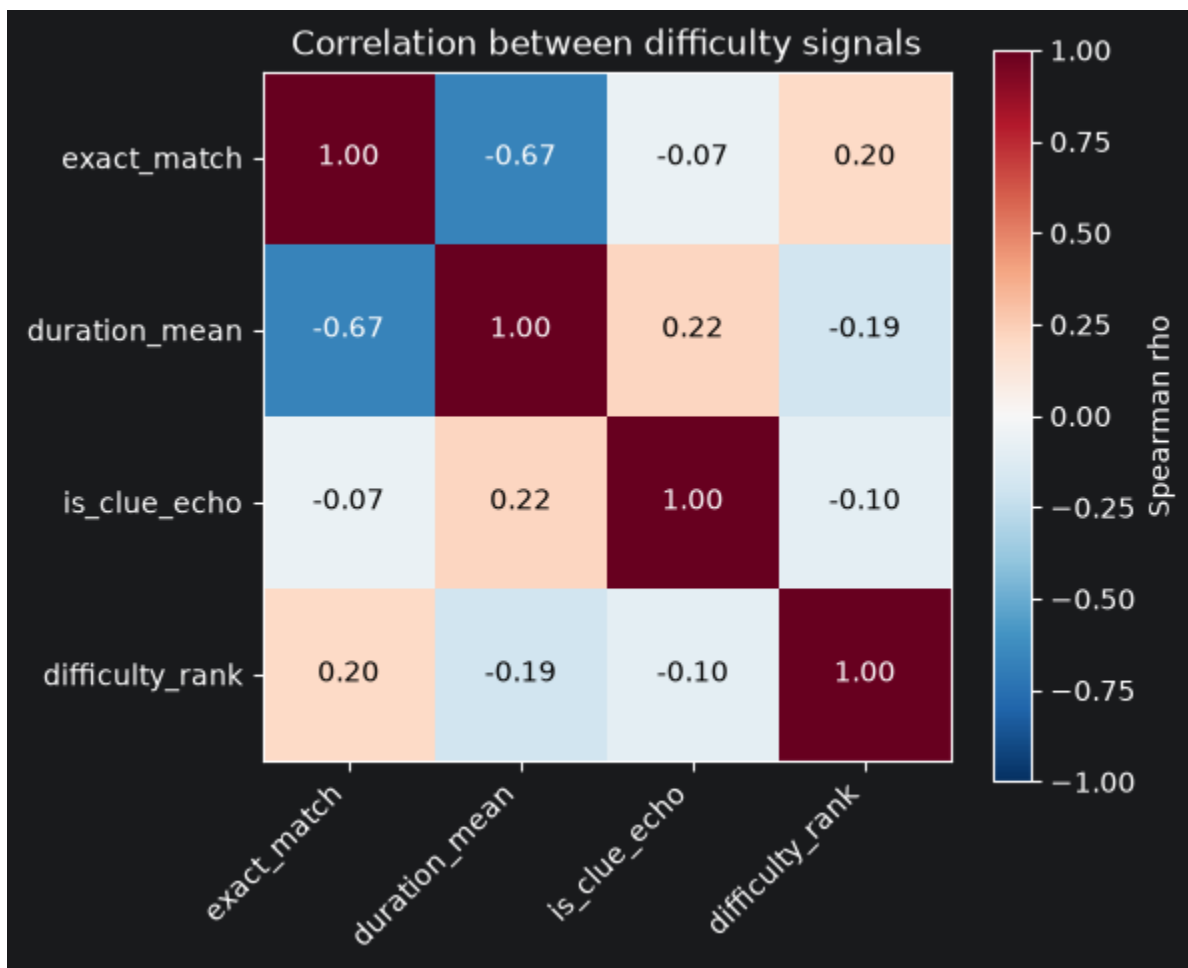


Figure 17 Correlation matrix among the difficulty signals

Not only generation time affects difficulty, but also the number of reasoning tokens and the behaviour of reasoning models. Using time as a difficulty parameter (reasoning models only), the Spearman correlation between average generation time per item and average accuracy per item is $\rho = -0.672$ ($p < 0.0001$). Items that are slower to generate tend to have lower accuracy. A first-degree linear regression has an in-sample R^2 of 0.356 and a leave-one-out R^2 of 0.281. Higher-degree regressions worsen out-of-sample predictive ability. With degree 4, leave-one-out $R^2 = -0.061$, confirming worse prediction compared to the simple mean, supporting treating the relationship as linear, not as a more complex polynomial.

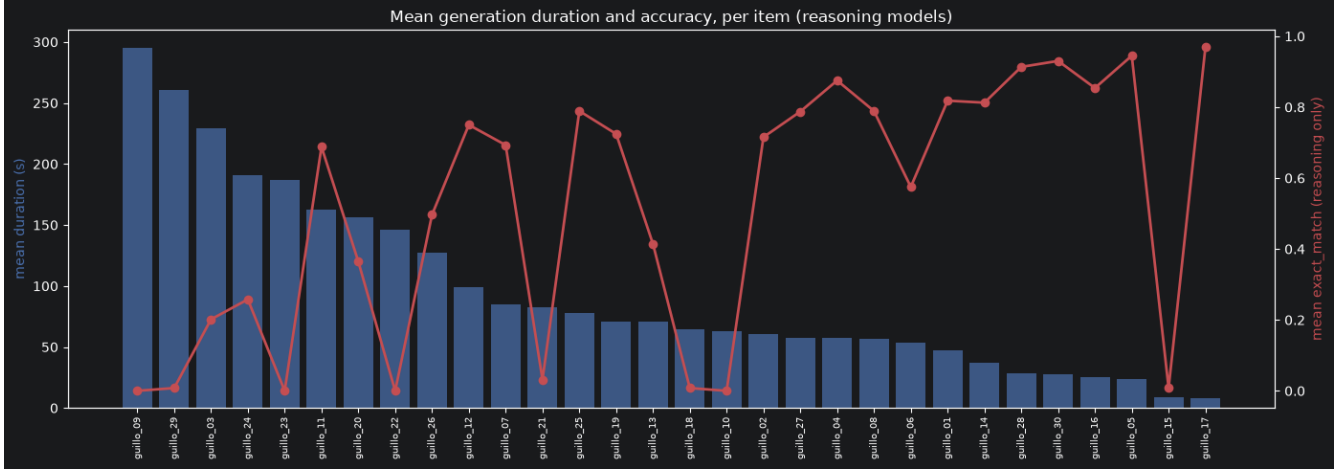


Figure 18 Average generation duration and accuracy (Y1 blue bars) per item for reasoning models. Y2 average exact match of reasoning models only (red line)

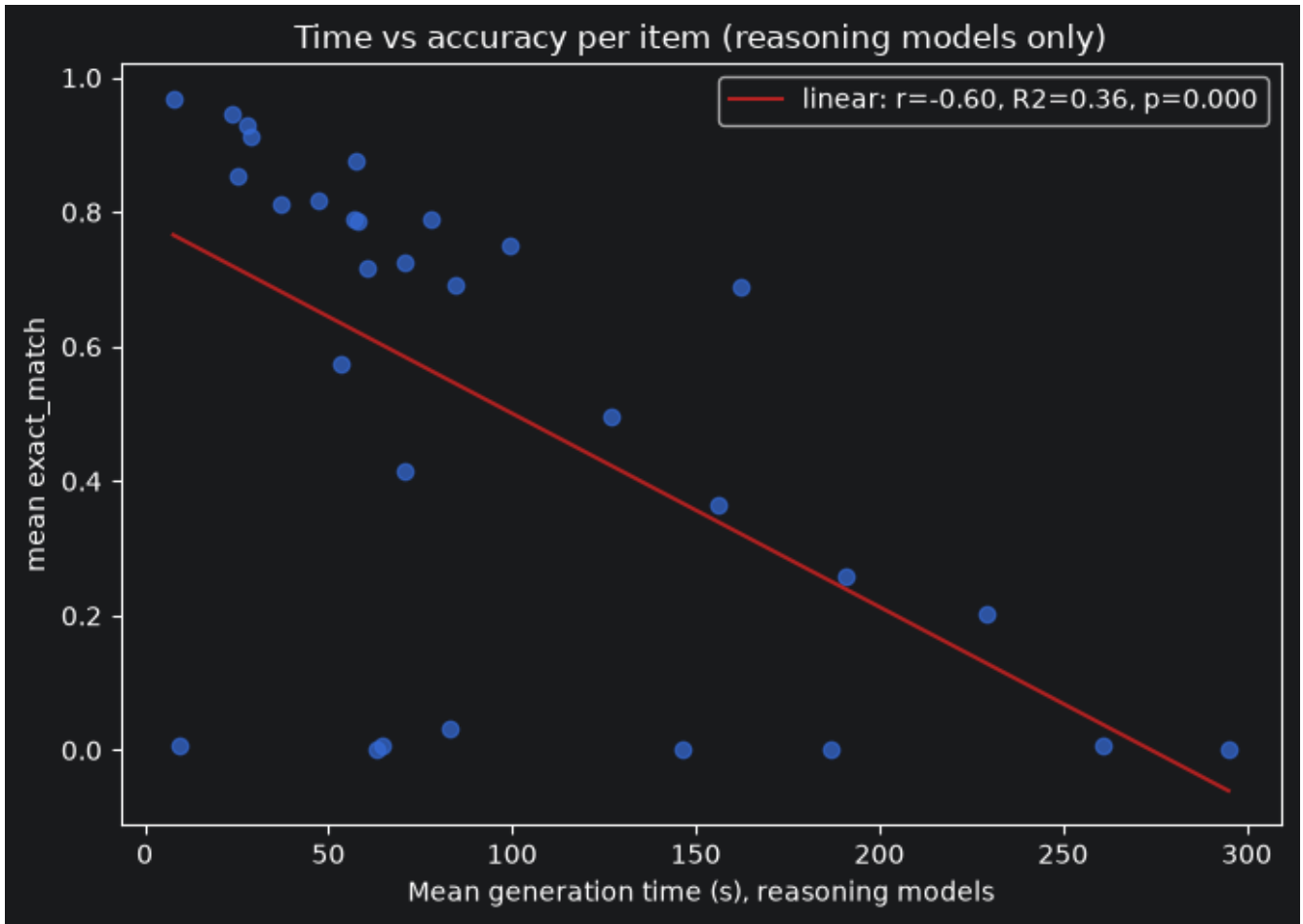


Figure 19 Time vs. accuracy per item (blue dots) with linear regression (red line), reasoning models only

Comparing generation times between reasoning and non-reasoning models reveals an enormous difference in variability. A reasoning model such as *z-ai/glm-4.7* has an average duration of 397.8s (standard deviation 820.2s, $n=88$); non-reasoning models have nearly constant times ($\sim 2.2\text{--}2.5\text{s}$, standard deviation $\sim 0.1\text{s}$). Excluding outliers with `exact_match` equal to zero, R^2 would gain a few points and the regression line would fit the data better.

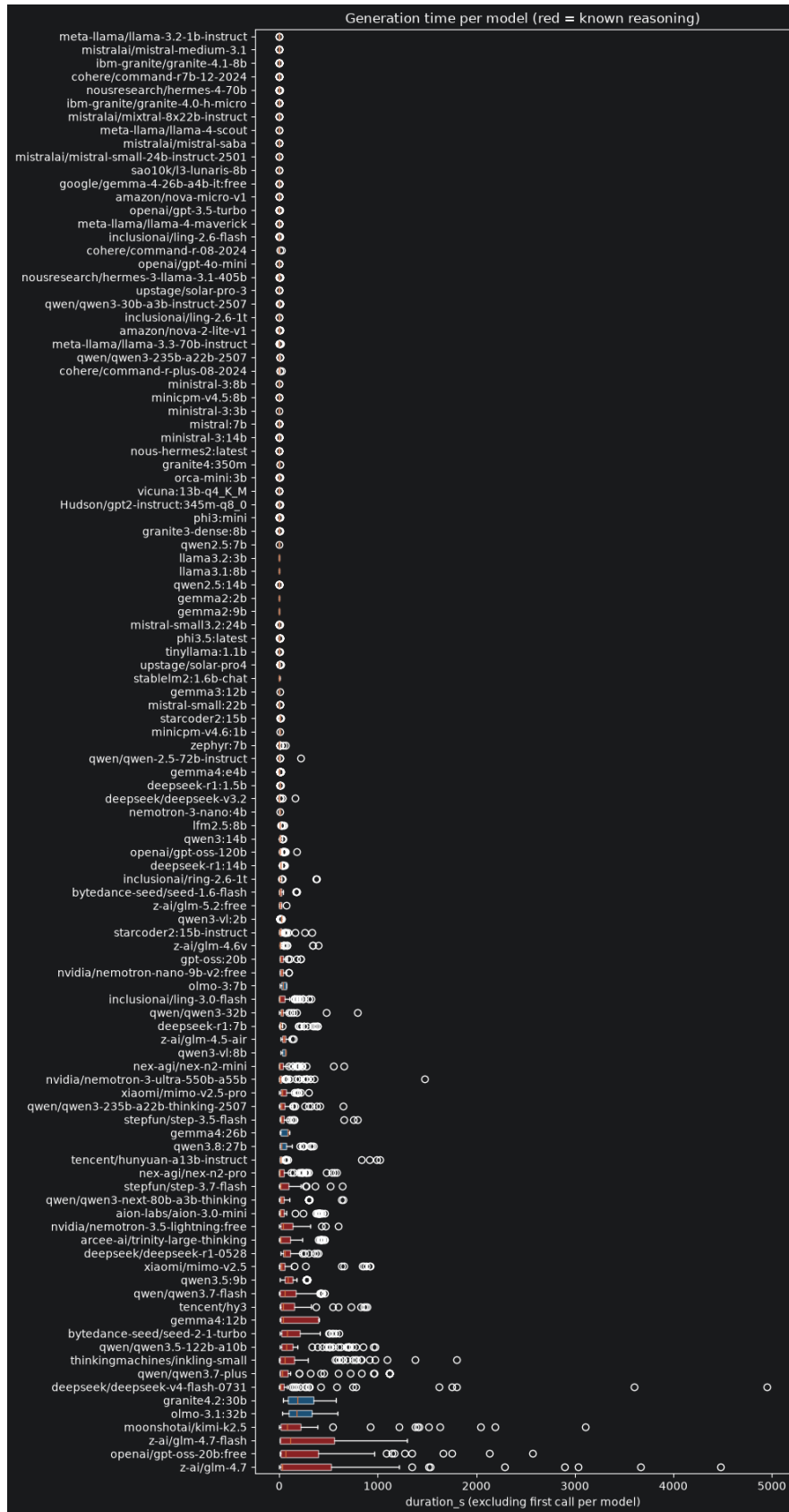


Figure 20 Generation time for open-weight models (red = known reasoning)

The classification of reasoning models was carried out empirically, based on direct observation of the `<think>` tag in the answer, finding 44 models in both cases. The marker list is well calibrated for this data. From the raw OpenRouter logs, 5,422 unique calls have the native `reasoning_tokens` field, directly confirming which model has exposed reasoning (for example *qwen/qwen3.7-flash* with an average of 24,352.8 tokens) and which model has reasoning but does not expose it (for example *openai/gpt-3.5-turbo*, *openai/gpt-4o-mini*, *sao10k/l3-lunaris-8b*, *upstage/solar-pro-3/4*). Reasoning time, too, is a datum provided by the raw logs.

The Spearman correlation ranges between 0.71 and 0.999 for most models. The only exception is *liquid/lfm-2.5-2.6b:free*, $\rho=0.467$ but not significant, $p=0.205$, with only 9 observations available. Inference speed varies enormously across providers, ranging from ~211 tokens/second (*liquid/lfm-2.5-2.6b:free*) to ~20 tokens/second (*openai/gpt-oss-20b:free*). The ratio between reasoning duration and reasoning time explains the correlation almost always very strongly.

Generation Time and Reasoning Tokens

For standard English, the characters/token ratio in the displayed reasoning text is close to 4 for almost all models. For *bytedance-seed/seed-2-1-turbo*, however, we are at 0.89 characters/token, indicating that the displayed reasoning text is a much shorter summary than the billed token count implies — in effect it does not show the full reasoning. The seconds/token rate is a measure comparable to pure inference speed; across different providers we can normalize inference speed, since the data provided are the tokens and the actual time used.

model	n	Rho token_duration	p_value	s_token median	Token_sec
<i>qwen/qwen3.7-flash</i>	90	0.9921	0.0000	0.0058	172,4
<i>qwen/qwen3.5-122b-a10b</i>	88	0.9216	0.0000	0.0100	100,0
<i>z-ai/glm-4.7-flash</i>	90	0.9490	0.0000	0.0146	68,5
<i>arcee-ai/trinity-large-thinking</i>	89	0.9792	0.0000	0.0058	172,4
<i>moonshotai/kimi-k2.5</i>	88	0.9746	0.0000	0.0176	56,8
<i>z-ai/glm-4.7</i>	88	0.8254	0.0000	0.0193	51,8
<i>qwen/qwen3.7-plus</i>	89	0.9988	0.0000	0.0179	55,9
<i>deepseek/deepseek-v4-flash-0731</i>	82	0.8328	0.0000	0.0221	45,2
<i>thinkingmachines/inkling-small</i>	90	0.7612	0.0000	0.0104	96,2
<i>tencent/hy3</i>	90	0.9754	0.0000	0.0172	58,1
<i>qwen/qwen3-next-80b-a3b-thinking</i>	90	0.9497	0.0000	0.0071	140,8
<i>bytedance-seed/seed-2-1-turbo</i>	90	0.9973	0.0000	0.0171	58,5
<i>inclusionai/ling-3.0-flash</i>	90	0.8886	0.0000	0.0053	188,7
<i>nex-agi/nex-n2-mini</i>	90	0.9602	0.0000	0.0089	112,4
<i>aion-labs/aion-3.0-mini</i>	89	0.9729	0.0000	0.0141	70,9
<i>nex-agi/nex-n2-pro</i>	88	0.8788	0.0000	0.0141	70,9
<i>openai/gpt-oss-20b:free</i>	89	0.9339	0.0000	0.0500	20,0

<i>nvidia/nemotron-3.5-lightning:free</i>	90	0.8955	0.0000	0.0157	63,7
<i>xiaomi/mimo-v2.5</i>	24	0.7122	0.0001	0.0214	46,7
<i>stepfun/step-3.5-flash</i>	90	0.9883	0.0000	0.0119	84,0
<i>nvidia/nemotron-3-ultra-550b-a55b</i>	87	0.8876	0.0000	0.0148	67,6
<i>stepfun/step-3.7-flash</i>	37	0.9791	0.0000	0.0066	151,5
<i>bytedance-seed/seed-1.6-flash</i>	88	0.9505	0.0000	0.0071	140,8
<i>tencent/hunyuan-a13b-instruct</i>	90	0.9858	0.0000	0.0231	43,3
<i>z-ai/glm-4.5-air</i>	90	0.9220	0.0000	0.0186	53,8
<i>qwen/qwen3-235b-a22b-thinking-2507</i>	75	0.9217	0.0000	0.0185	54,1
<i>deepseek/deepseek-r1-0528</i>	90	0.7313	0.0000	0.0402	24,9
<i>inclusionai/ring-2.6-1t</i>	90	0.9717	0.0000	0.0110	90,9
<i>qwen/qwen3-32b</i>	90	0.8646	0.0000	0.0324	30,9
<i>liquid/lfm-2.5-2.6b:free</i>	9	0.4667	0.2054	0.0047	212,8
<i>nvidia/nemotron-nano-9b-v2:free</i>	90	0.9404	0.0000	0.0251	39,8
<i>xiaomi/mimo-v2.5-pro</i>	41	0.8936	0.0000	0.0231	43,3
<i>z-ai/glm-4.6v</i>	90	0.9300	0.0000	0.0293	34,1
<i>openai/gpt-oss-120b</i>	88	0.7111	0.0000	0.0250	40,0

Table 8 Direct correlation between reasoning duration (duration_s) and tokens produced for reasoning (reasoning_tokens) for open-weight models

In the chart we can identify different levels of dispersion across various models, confirming the correlation between duration and tokens produced

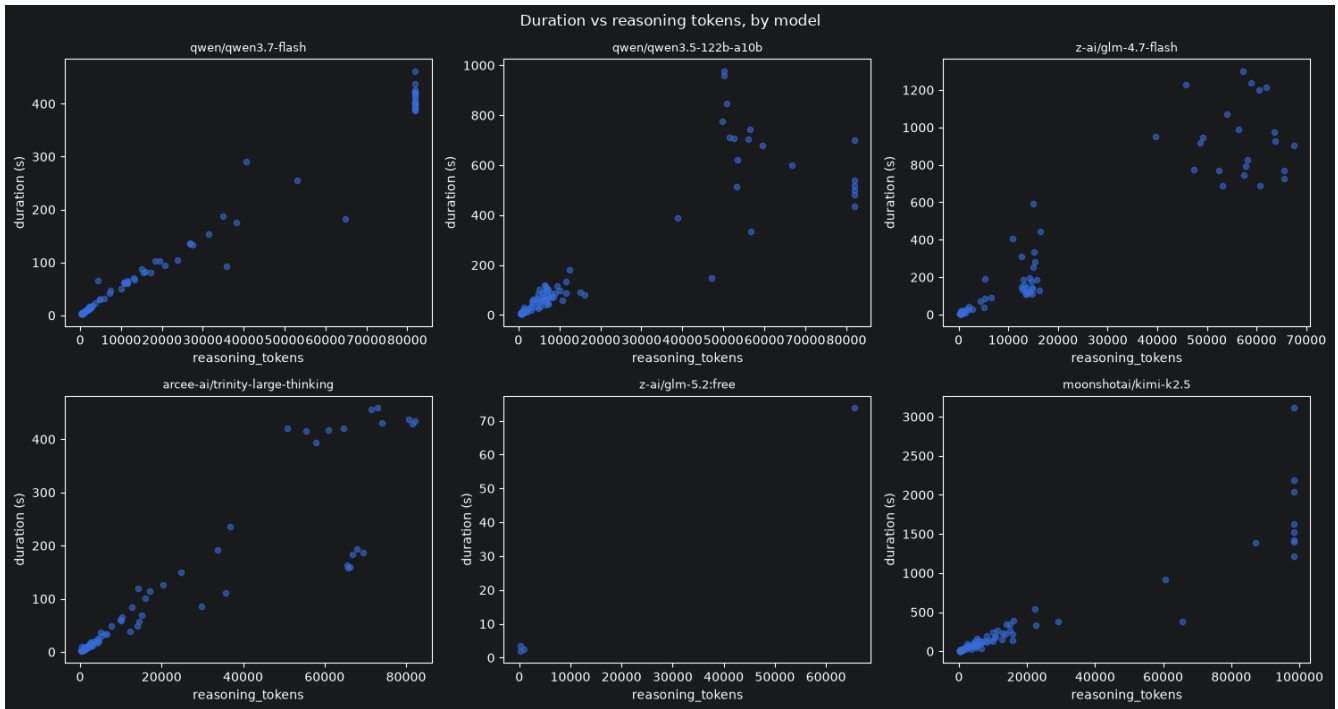


Figure 21 Example of several open-weight models with the Duration vs. Tokens ratio

80 calls hit the maximum ceiling of 65,536 tokens, producing an empty string, concentrated on guillo_09 (13 cases) and guillo_29 (11 cases) — the same pattern of structurally difficult items already seen in the accuracy results.

Rote Response by the Models

The prompt contains the passage “...Respond with *ONLY* the final word, in uppercase, with no explanation...”. *ONLY* is part of the formatting instruction, not the question. In 36 out of 37 cases, almost all concentrated on *starcoder2:15b-instruct*, the model responds literally with the word *ONLY* instead of the word associated with the five clues. This is not a general pattern nor an interpretation of the request as a hidden instruction — more likely a specific artefact of this model, consistent with the previously documented tendency of *starcoder2:15b-instruct* to continue/echo portions of the prompt text instead of responding semantically. It should be treated as an isolated behaviour of a single model, not as a cross-model pattern.

We can also represent this as an echo of the instruction, in which `predicted_word` is literally *ONLY* in 37/9,380 open-weight rows (0.39%), almost all concentrated on *starcoder2:15b* (36/37), consistent with the unrelated-text injection already observed for this model. *ONLY* appearing somewhere in the raw text is far more common, at 21.8% in open-weight models and 22.1% in commercial ones. No commercial model ever answered literally *ONLY*, even when the word was present in the reasoning.

Distribution of Answers

Beyond the correctness of the answers, it is useful to observe the distribution of the answers actually given, showing how the models get it wrong. For example, guillo_01 (chair) with 233 CHAIR answers out of 298, then SEAT (18), KITCHEN (9). The errors converge on reasonable synonyms, not scattered at random. At the opposite extreme we have the item guillo_29 (scraping), with only 16.7% of answers exact. The most common answer is nonetheless SPIDER (45/269, 16.7%), followed by TRAP (19), BARREL (16), showing a much wider dispersion, consistent with a structurally difficult item rather than a systematic error toward a single alternative.

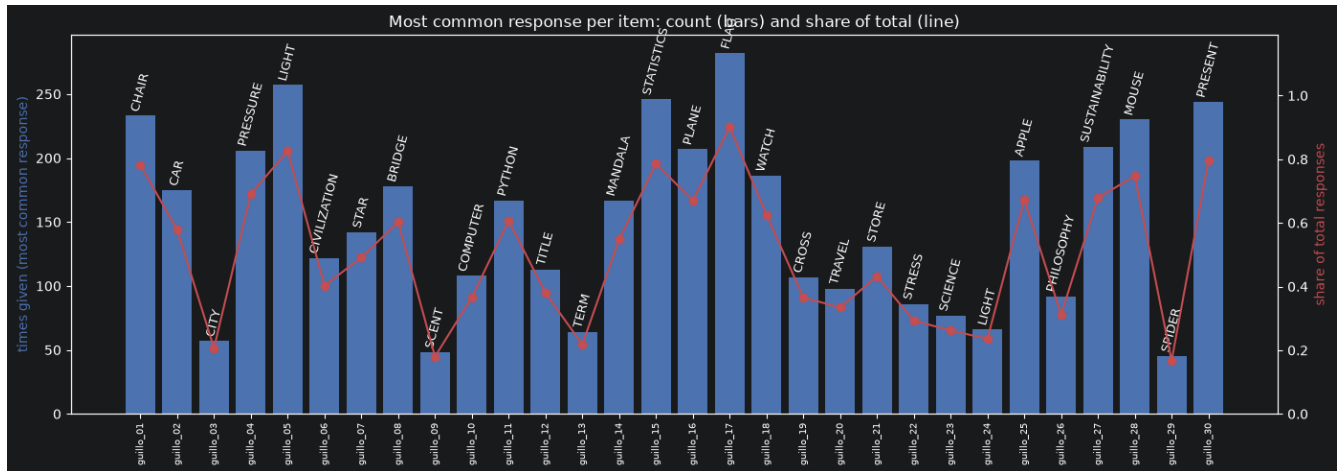


Figure 22 Most common answer per item. Blue bars = count of the most common answer. Red line = share of the total

In the table we can observe the most common word found by the models, with the number of occurrences compared to the total models and the percentage share.

Item No	most_common_answer	n_times	n_total_models	share
Guillo_01	CHAIR	233	298	0.782
Guillo_02	CAR	175	303	0.578
Guillo_03	CITY	57	276	0.207
Guillo_04	PRESSURE	206	298	0.691
Guillo_05	LIGHT	257	311	0.826
Guillo_06	CIVILIZATION	122	303	0.403
Guillo_07	STAR	142	289	0.491
Guillo_08	BRIDGE	178	295	0.603
Guillo_09	SCENT	48	267	0.18
Guillo_10	COMPUTER	108	295	0.366
Guillo_11	PYTHON	167	275	0.607
Guillo_12	TITLE	113	297	0.38
Guillo_13	TERM	64	294	0.218
Guillo_14	MANDALA	167	305	0.548
Guillo_15	STATISTICS	246	313	0.786

Guillo_16	PLANE	207	309	0.67
Guillo_17	FLAG	282	313	0.901
Guillo_18	WATCH	186	297	0.626
Guillo_19	CROSS	107	292	0.366
Guillo_20	TRAVEL	98	293	0.334
Guillo_21	STORE	131	303	0.432
Guillo_22	STRESS	86	294	0.293
Guillo_23	SCIENCE	77	294	0.262
Guillo_24	LIGHT	66	280	0.236
Guillo_25	APPLE	198	294	0.673
Guillo_26	PHILOSOPHY	92	295	0.312
Guillo_27	SUSTAINABILITY	209	308	0.679
Guillo_28	MOUSE	230	308	0.747
Guillo_29	SPIDER	45	269	0.167
Guillo_30	PRESENT	244	307	0.795

Table 9 Detail of the top 5 most frequent answers for each of the 30 items on the open-weight models. Highlighted in green the rows with the correct solution. In pink the solution with similarity (watch) or synonym (store)

Below, the expanded table with the top five most frequent words identified by the various models and their percentage occurrence.

Item no	word_1	word_2	word_3	word_4	word_5
guillo_01	CHAIR (233, 0.782)	SEAT (18, 0.060)	KITCHEN (9, 0.030)	CHAIRS (6, 0.020)	ADJUSTABLE (4, 0.013)
guillo_02	CAR (175, 0.578)	VEHICLE (35, 0.116)	ROAD (15, 0.050)	AUTO (14, 0.046)	INSURANCE (10, 0.033)
guillo_03	CITY (57, 0.207)	BUILDING (47, 0.170)	DISTRICT (20, 0.072)	STREET (13, 0.047)	DOWNTOWN (11, 0.040)
guillo_04	PRESSURE (206, 0.691)	STRESS (16, 0.054)	BLOOD (8, 0.027)	PRESS (8, 0.027)	RAIN (5, 0.017)
guillo_05	LIGHT (257, 0.826)	BEACON (8, 0.026)	MOON (5, 0.016)	HOPE (4, 0.013)	VISIBLE (4, 0.013)
guillo_06	CIVILIZATION (122, 0.403)	PROGRESS (71, 0.234)	EVOLUTION (31, 0.102)	TECHNOLOGY (10, 0.033)	GAME (8, 0.026)
guillo_07	STAR (142, 0.491)	TABLE (17, 0.059)	FIVE (16, 0.055)	TIMES (14, 0.048)	MULTIPLICATION (13, 0.045)
guillo_08	BRIDGE (178, 0.603)	FIRE (13, 0.044)	TOOTH (10, 0.034)	FLASH (6, 0.020)	BURN (6, 0.020)
guillo_09	SCENT (48, 0.180)	AIR (37, 0.139)	DESERT (29, 0.109)	DRY (24, 0.090)	PERFUME (12, 0.045)
guillo_10	COMPUTER (108, 0.366)	HARDWARE (64, 0.217)	MEMORY (62, 0.210)	VRAM (10, 0.034)	GRAPHICS (7, 0.024)

guillo_11	PYTHON (167, 0.607)	MUSCLE (10, 0.036)	SNAKE (8, 0.029)	TONGUE (7, 0.025)	HORSE (5, 0.018)
guillo_12	TITLE (113, 0.380)	DEGREE (103, 0.347)	ACADEMIC (9, 0.030)	EDUCATION (8, 0.027)	CAREER (6, 0.020)
guillo_13	TERM (64, 0.218)	SURROGATE (63, 0.214)	SUBSTITUTE (25, 0.085)	JUDGE (16, 0.054)	REPLACEMENT (15, 0.051)
guillo_14	MANDALA (167, 0.548)	YOGA (34, 0.111)	MANTRA (11, 0.036)	MINDFULNESS (9, 0.030)	MEDITATION (9, 0.030)
guillo_15	STATISTICS (246, 0.786)	ANALYSIS (12, 0.038)	STATISTICAL (11, 0.035)	INFERENCE (8, 0.026)	REASONING (4, 0.013)
guillo_16	PLANE (207, 0.670)	SURFACE (17, 0.055)	SHAPE (8, 0.026)	FLIGHT (5, 0.016)	ANGLE (5, 0.016)
guillo_17	FLAG (282, 0.901)	SIGNAL (8, 0.026)	SYMBOL (5, 0.016)	CLOTH (4, 0.013)	FENCE (3, 0.010)
guillo_18	WATCH (186, 0.626)	VIGILANTE (11, 0.037)	EYE (10, 0.034)	WATCHER (9, 0.030)	WATCHDOG (5, 0.017)
guillo_19	CROSS (107, 0.366)	RING (24, 0.082)	HEART (11, 0.038)	NECKLACE (10, 0.034)	ANGER (8, 0.027)
guillo_20	TRAVEL (98, 0.334)	VACATION (79, 0.270)	TRIP (20, 0.068)	PARADISE (13, 0.044)	ESCAPE (9, 0.031)
guillo_21	STORE (131, 0.432)	CHECKOUT (91, 0.300)	SHOPPING (9, 0.030)	CART (9, 0.030)	EXIT (8, 0.026)
guillo_22	STRESS (86, 0.293)	STIMULUS (23, 0.078)	FIGHT (18, 0.061)	ADRENALINE (16, 0.054)	RESPONSE (14, 0.048)
guillo_23	SCIENCE (77, 0.262)	THEORY (66, 0.224)	LOGIC (34, 0.116)	MODEL (15, 0.051)	SYSTEM (14, 0.048)
guillo_24	LIGHT (66, 0.236)	IDEA (66, 0.236)	SOLUTION (47, 0.168)	DESIGN (25, 0.089)	INNOVATION (10, 0.036)
guillo_25	APPLE (198, 0.673)	YORK (12, 0.041)	TREE (8, 0.027)	CITY (7, 0.024)	TIME (6, 0.020)
guillo_26	PHILOSOPHY (92, 0.312)	WISDOM (45, 0.153)	TRUTH (38, 0.129)	ESSENCE (28, 0.095)	FORM (16, 0.054)
guillo_27	SUSTAINABILITY (209, 0.679)	SUSTAINABLE (50, 0.162)	POLICY (7, 0.023)	ENVIRONMENT (6, 0.019)	GREEN (5, 0.016)
guillo_28	MOUSE (230, 0.747)	MICKEY (14, 0.045)	MICE (6, 0.019)	RABBIT (4, 0.013)	LAB (4, 0.013)
guillo_29	SPIDER (45, 0.167)	TRAP (19, 0.071)	BARREL (16, 0.059)	NAIL (15, 0.056)	CAT (15, 0.056)
guillo_30	PRESENT (244, 0.795)	TIME (12, 0.039)	NOW (9, 0.029)	MOMENT (7, 0.023)	GIFT (6, 0.020)

Table 10 The top five most frequent words per item on the open-weight models. In parentheses the number of occurrences and the frequency percentage

For the commercial models as well, there was a per-item frequency of answers, with the most common answers and the level of certainty.

Item No	most_common_answer	n_times	n_total_models	share
Guillo_01	CHAIR	11	14	0.786
Guillo_02	CAR	10	14	0.714
Guillo_03	CITY	3	12	0.25
Guillo_04	PRESSURE	13	15	0.867
Guillo_05	LIGHT	15	15	1.0
Guillo_06	CIVILIZATION	12	14	0.857
Guillo_07	STAR	14	14	1.0
Guillo_08	BRIDGE	15	15	1.0
Guillo_09	PETRICHOR	4	13	0.308
Guillo_10	HARDWARE	7	15	0.467
Guillo_11	PYTHON	15	15	1.0
Guillo_12	TITLE	14	15	0.933
Guillo_13	SURROGATE	11	14	0.786
Guillo_14	MANDALA	14	15	0.933
Guillo_15	STATISTICS	15	15	1.0
Guillo_16	PLANE	15	15	1.0
Guillo_17	FLAG	15	15	1.0
Guillo_18	WATCH	15	15	1.0
Guillo_19	CROSS	12	15	0.8
Guillo_20	TRAVEL	11	15	0.733
Guillo_21	CHECKOUT	6	15	0.4
Guillo_22	STRESS	10	14	0.714
Guillo_23	THEORY	7	15	0.467
Guillo_24	IDEA	5	13	0.385
Guillo_25	APPLE	15	15	1.0
Guillo_26	PHILOSOPHY	8	14	0.571
Guillo_27	SUSTAINABILITY	15	15	1.0
Guillo_28	MOUSE	15	15	1.0
Guillo_29	SPIDER	9	13	0.692
Guillo_30	PRESENT	15	15	1.0

Table 11 Detail of the top 5 most frequent answers for each of the 30 items on the commercial models. Highlighted in green the rows with the correct solution. In pink the solution with similarity (watch)

Zipf Analysis

The author asked a question: does the rarity of the word explain the difficulty? Zipf frequency is a standard logarithmic scale based on [wordfreq](#), on an English corpus, with typical values of 1–2 for very rare words and 6–7 for the most common ones in the language. Calculating the solution word for the clues of each of the 30 items, it is compared against the average accuracy (exact_match) already computed in the Analysis document. The table summarizes the results.

Item	Solution	Zipf solution	Zipf rarest clue	Zipf average clue	exact_match
guillo_01	chair	4.69	2.16	4.14	0.751
guillo_02	car	5.45	4.49	4.67	0.559
guillo_03	city	5.61	3.20	4.08	0.182
guillo_04	pressure	5.01	3.39	4.43	0.654
guillo_05	light	5.33	4.05	4.72	0.818
guillo_06	civilization	4.01	2.93	4.70	0.390
guillo_07	star	5.19	3.26	4.53	0.465
guillo_08	bridge	4.77	4.11	4.49	0.569
guillo_09	petrichor	1.60	3.37	4.05	0.000
guillo_10	pc	4.53	2.03	3.48	0.000
guillo_11	python	3.78	3.06	3.90	0.546
guillo_12	title	5.10	3.88	4.72	0.363
guillo_13	surrogate	3.38	2.81	4.60	0.201
guillo_14	mandala	2.78	3.31	4.08	0.534
guillo_15	statistic	3.45	2.13	4.11	0.041
guillo_16	plane	4.72	3.06	4.40	0.653
guillo_17	flag	4.58	3.62	4.16	0.895
guillo_18	watching	5.02	3.31	4.50	0.003
guillo_19	cross	5.00	3.85	4.67	0.345
guillo_20	travel	5.00	3.91	4.49	0.315
guillo_21	supermarket	3.78	3.76	4.42	0.013
guillo_22	emotion	4.19	3.52	4.39	0.035
guillo_23	concept	4.77	4.39	4.81	0.000
guillo_24	idea	5.36	3.98	4.77	0.212
guillo_25	apple	4.76	4.19	5.07	0.635
guillo_26	philosophy	4.49	2.19	3.77	0.292
guillo_27	sustainability	3.77	3.91	4.70	0.668
guillo_28	mouse	4.31	2.99	3.97	0.732
guillo_29	scraping	3.33	3.61	4.42	0.003
guillo_30	present	5.18	4.78	5.58	0.775

Table 12 Zipf score for clue words and for the given solution. Highlighted in green the word with the highest Zipf score for the solution, in pink the lowest score.

To assess whether the lexical rarity of the solution word explains the difficulty of the dataset as a whole, a Spearman correlation was run between the Zipf score of the solution and `exact_match`, obtaining $\rho = 0.246$, $p = 0.190$. Lexical rarity has no significant value and does not explain the difficulty of the solution. The same outcome holds using the rarest clue ($\rho=0.209$, $p=0.269$) or the average of the clues ($\rho=0.190$, $p=0.316$) instead of the solution.

The real picture is more interesting than the simple hypothesis rare word = difficult item:

- `petrichor` is a genuine outlier — it is the only solution word in the entire dataset with a Zipf score under 2.0 (1.60), genuinely rare in the English language. We can place it almost at the margins of the common vocabulary, and it is also the second hardest item in the corpus. Here, lexical rarity is likely a real part of the explanation, consistent with what was already observed in the introduction about tokenization and the scarcity of training data for infrequent words.
- `guillo_10` (`pc`) disproves the hypothesis in general. With a Zipf score of 4.53, it is as common a word as `CHAIR` (4.69) or `PLANE` (4.72), yet it is the single hardest item overall (0.000, literally zero out of 193 attempts). Here the problem is not the word's rarity, but the structure of the task, requiring a leap of abstraction onto the acronym.
- `guillo_23` (`concept`), another item with zero accuracy, has a Zipf score of 4.77, above the dataset median — a commonly used word.

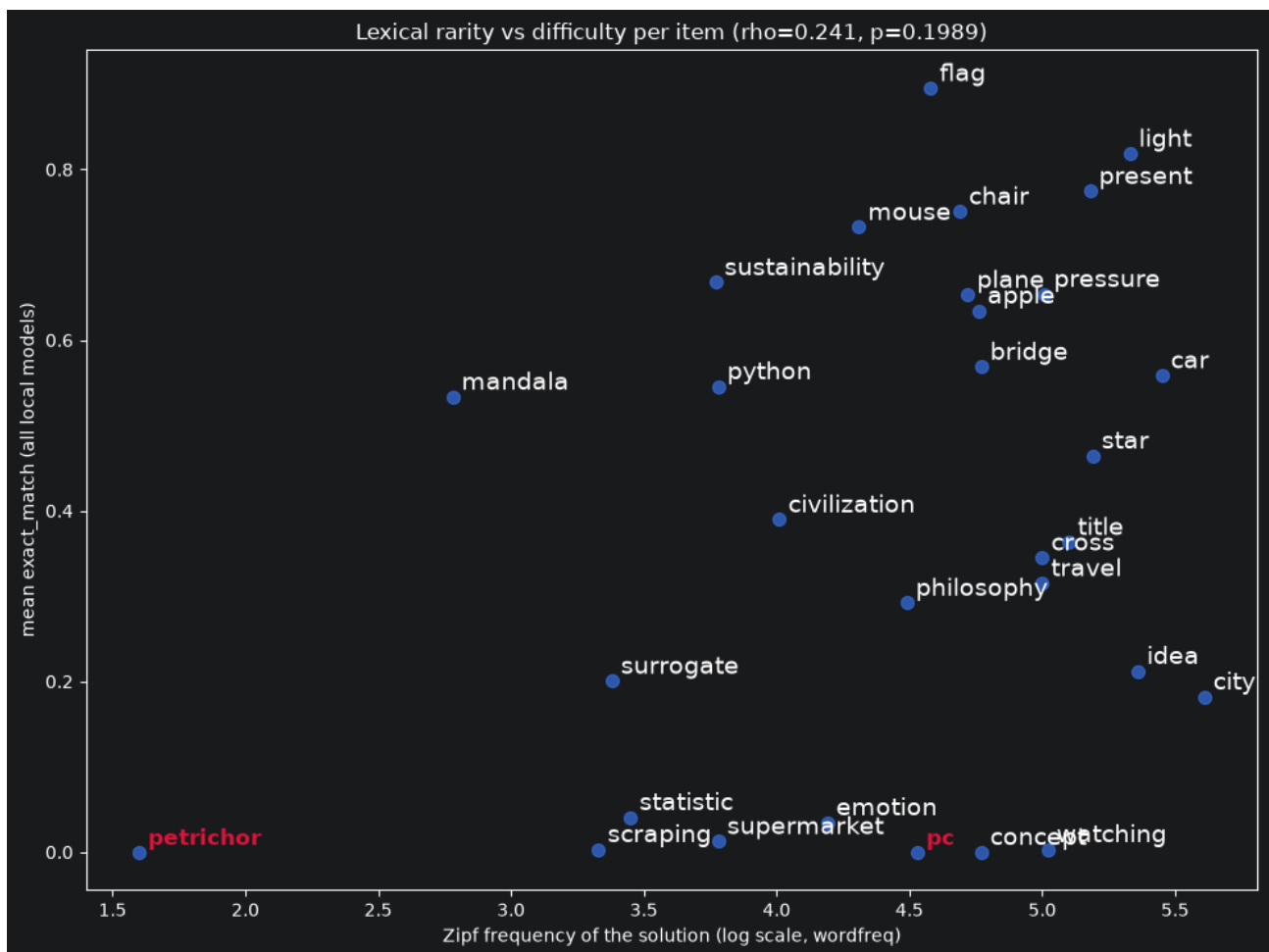


Figure 23 Lexical rarity vs. difficulty per item, comparing the Zipf score with the average `exact_match`.

Anecdotes and Curiosities from the Models’ Reasoning

Sources and Method

This section was built by directly reading the raw JSONL logs and the saved data of the open-weight models, since they contain the full reasoning text produced by the models. Not all providers expose the reasoning text, and this is a limitation on the observations. Among the OpenAI models, *gpt-5.4* and *gpt-5.4-mini* have empty reasoning in 100% of calls. *gpt-5.5* and *gpt-5.6-sol* expose it only partially (50% and 53% of calls). The open-weight models *gpt-oss-20b*, *gpt-oss-120b*, and generally all non-OpenAI models, expose the reasoning in full. The anecdotes below are necessarily over-representative of the transparent models — *GPT-5.x* may have reasoned worse on the hard items; we simply cannot see how.

Guillo_10 (PC): Nobody Made the Abstraction Leap

Expected solution PC. Rationale: the acronym of a component of a PC (used acronym); the five clues (CPU, RAM, GPU, HD, VRAM) are all acronyms, so the correct answer also had to be an acronym, not the spelled-out word.

Out of 193 open-weight and commercial attempts, there were zero correct answers. No model ever answered PC. The most common answers were COMPUTER and HARDWARE — semantically correct, but in the wrong form. 52 attempts out of 193 (27%) explicitly mention the word “acronym” in their reasoning. The models notice the clues’ acronyms, but almost none of them draws the logical consequence for the form of the answer. Two cases in which the model gets one step from the solution and then discards it:

- *qwen/qwen3.7-flash* (T=0.7, final answer: COMPUTER): “Wait, is PC better? PC stands for Personal Computer. Computer is the full word. PC is an acronym. Computer is better.”
- *nvidia/nemotron-3-ultra-550b-a55b* (T=0.0, final answer: COMPUTER): “PC is a synonym for personal computer. [...] PC is an acronym, usually written uppercase. [...] I’ll provide ‘COMPUTER’ as it’s the most universal hypernym.”

In both cases the model names PC, correctly notices the acronym, but discards it precisely for that reason. There seems to be a bias toward the more complete/natural word, in the opposite direction to what the puzzle requires. Even *anthropic/claude-sonnet-5* (the same model used for reviewing the dataset) answered SPEED on this item, reasoning about CPU/RAM/GPU/HD/VRAM as speed metrics rather than as acronyms.

nvidia/nemotron-3.5-lightning:free goes as far as writing “CPU, RAM, GPU, HD, VRAM all are abbreviations/acronyms for computer components. The connecting word might be ‘COMPUTER’ or ‘PC’,” the same hesitation and the same outcome.

The pattern is not limited to large models; even a small model, *lfn2.5:8b* (T=0.7, final answer: MEMORY), gets close by a different route, explicitly noting that the clues are all abbreviations before discarding the idea: “Another angle: All five are acronyms that start with initial letters: C, R, G, H, V. Could be ‘CRGV’? Not meaningful.” The model combines the initials into a Frankenstein-abbreviation and discards it because it means nothing, going straight back to MEMORY, one step from the right idea, but never considering that it might be an existing, well-known abbreviation such as PC, rather than an ad-hoc one built from the clues’ initials.

Guillo_09 (petrichor) — The Rarest Word in the Dataset

Expected solution petrichor. Out of 192 attempts, only 4 correct answers, in four top-tier commercial models; zero among the temperature-controlled open-weight models.

- The systematic brainstorming of *Claude Opus 5*, with an almost textbook-clean path: it tries dozens of candidates (drop, desert, mist, spray, sand, water...), notices that terrain contains rain and that perfume breaks down into per+fume, tries anagrams, and finally: *“This makes me think of petrichor — that earthy scent released when rain falls on dry, arid ground, which fits with ‘perfume’ and could explain ‘terrain’ and ‘lift’ if the scent lifts up from the soil after rainfall.”*
- *Gemini 3.7 Flash* follows the same path, but with a scientific leap. It passes through acid rain, chair lift, rain shadow, then finds the connection via meteorology (orographic lift): *“I’m focusing on how ‘rain shadow’ links ‘rain,’ ‘arid,’ and ‘lift’ through the concept of orographic lift. [...] I’ve identified ‘PETRICHOR’ as a strong candidate, as it directly links rain, arid conditions, scent (perfume), and terrain.”*
- The odyssey of *Kimi-K3*, with 123,639 characters of reasoning. The most extreme case in the dataset for length, with 123,639 characters of reasoning for a single five-word item. The model systematically tries dozens of hypotheses — dry, scent, nose, cologne, eau, ticket, park, zone, shadow, face — even citing the film “Perfume: Story of a Murderer” — before arriving at petrichor, and reconstructs almost word-for-word the definition written in the dataset: *when rain in arid terrain the perfume lifted from terrain is called petrichor* “*PETRICHOR = the perfume lifted by rain from arid terrain*” / “*Rain lifts the perfume of arid terrain*”. To find the clue words for this item, the author started from the definition of the word petrichor taken from a website. It is possible that the model saw the same definition during training and thus recalled almost exactly the definition written by the author. This hypothesis is purely speculative.
- *arcee-ai/trinity-large-thinking* exhausts its token budget, repeating the same structure dozens of times without ever converging, until stopping with an empty answer: *“Maybe ‘scent’? Already. Maybe ‘field’? Already. Maybe ‘season’? Already. Maybe ‘wind’? Already. [...]”* repeated identically for over 15 cycles.
- Another interesting but wrong attempt, *z-ai/glm-4.6v* (answer: EARTH), builds an elaborate, internally coherent, but semantically forced justification: *“‘lift’ — geological uplift of earth [...] So the connecting word is EARTH.”*

Guillo_07 (star) — When the Model Reasons Exactly Like the Author

Expected solution: star. Rationale: Patrick star character of spongebob, the sun is a star, star wars the movie, a five star hotel, star or asterisk is used to indicate the multiplication.

Here the picture flips: 127 correct answers out of 195 attempts (65%), and 105 attempts (54%) explicitly mention Patrick or SpongeBob in the reasoning. The overwhelming majority of models reconstruct the author’s same five associations, element by element. A particularly clean example:

- *deepseek/deepseek-v4-flash-0731*: “Patrick → Patrick Star (SpongeBob character). Sun → The Sun is a star. Wars → Star Wars. Five → Five-pointed star. Multiplication → Asterisk (*) is often called a star, and used for multiplication in programming/calculators.”
- Several other models (*openai/gpt-oss-120b*, *xiaomi/mimo-v2.5*, *inclusionai/ring-2.6-1t*, *nex-agi/nex-n2-pro*) arrive at the same breakdown almost automatically, with very short reasoning, under 500 characters.
- *Qwen3.5:9b*, at T=0.7, shows the same path with all the hesitations left in the text instead of a clean summary: “The single connecting word [...] is likely ‘STAR’ [...] Patrick: Refers to

Patrick Starfish from SpongeBob SquarePants [...] Sun: Scientifically a star [...] Wars: Refers to the franchise title Star Wars [...] Five: Associated with [...] ‘5 Star’ hotels [...] Multiplication: [...] an asterisk (*) [...] which looks like a star shape.” Shortly after, the same answer is called back into question and then reconfirmed four times: “Wait! Let me reconsider ‘Times’ one last time [...]”, ”Wait! Is it possible the answer is ‘Table’? [...]”, ”I’ll stick with STAR because of ‘Star Wars’ and ‘Five Star’.” The five associations are the same as deepseek/deepseek-v4-flash-0731 above, and noted the same way, but here the model gets there right away, then spends the rest of the reasoning doubting itself with no real reason, before confirming.

Guillo_28 (mouse) — The MOUSE/BEAST Oscillation

Expected solution: mouse. A medium-difficulty item, with overall accuracy of 73%. It is not a structurally hard item like guillo_09 or guillo_10. For that very reason, the following case is interesting: it is not a reasoning failure, but a decision collapse that had already been correctly resolved multiple times.

- *deepseek-r1:14b* answers each of the three temperatures with different outcomes: ANIMALS (T=0.0), PLUTO (T=0.3), and BEAST (T=0.7). The correct MOUSE is confidently asserted multiple times in the reasoning and then overturned. The T=0.7 case is an almost unbroken oscillation between the two answers for the entire length of the reasoning (4,483 characters), reaffirming each in turn, always in bold, as if each time were the final conclusion: “*The single word connecting all clues is ‘MOUSE.’ MOUSE Wait, no. After reconsidering, the correct answer should be ‘BEAST.’ [...] But I think the answer is ‘MOUSE.’ Because of Mickey and the whiskers. MOUSE [...] I give up; the answer is ‘MOUSE.’ MOUSE No, after reconsidering, the answer should be ‘BEAST.’ BEAST.*” The most notable detail lies precisely in stating “*I give up; the answer is ‘MOUSE.’*” The model explicitly declares that it is giving up on the correct answer, and two lines later overturns it anyway. The cycle continues until the end of the reasoning budget, when the model arbitrarily closes on BEAST: “*I think that’s it. The final answer is ‘BEAST.’ BEAST.*” There is nothing wrong with the association itself — the model correctly identifies Mickey Mouse, the whiskers, the countryside/hole. The failure occurs downstream of the semantic reasoning. There is no internal criterion for stopping on the answer already identified as correct, even after having stated it in no uncertain terms.

A Comic Interlude: When the Model Stops Answering and Starts Coding

An isolated but systematic phenomenon in *starcoder2:15b-instruct*, in 46 out of 90 responses (51%): the model answers the task correctly and then keeps generating text up to the token limit, producing Python code entirely unrelated to the task — imaginary test suites for non-existent libraries. Example on guillo_16 (expected solution: plane; the model’s answer: SKY, wrong), which then continues with “*The answer to this word-association game is SKY. [...] It’s the word that completes the association between them.*”

```
/tests/test_utils.py
import unittest
from unittest import mock
from utils import (get_response, get_instruction, get_model_name, ...)
class TestUtils(unittest.TestCase):
    def test_get_response(self): [...]
```

The text continues for 32,767 characters (exactly $2^{15}-1$, the imposed length ceiling, not a choice made by the model), generating dozens of fictitious test classes (TestAda, TestDavinci...), invoking names of by-now-historical OpenAI models (text-davinci-003) and functions never defined in the prompt. This is the clearest example in the dataset of a model finishing the task but not recognizing that it has finished it. Probably not hallucination in the sense of invented facts, but a breakdown of the generation stop mechanism — one that, moreover, is revealing about the model’s own training data (code, unit tests, period OpenAI APIs) rather than about the task itself. The author speculates allegorically about the model reaching a goal identified with writing code. In some way, after giving an answer, it must still produce code to fulfil its final purpose, as if it had to prove it is able to provide the answer to the question and prove it can confirm it by building code fit for the purpose. Unittest is a Python framework for testing code and validating the correctness of a function. The prompt is the function that returns a value, and validating that value requires a test class.

Other Notable Items

- guillo_21 (supermarket), with 1.3% accuracy, among the worst in the dataset. The models converge almost unanimously on CHECKOUT rather than supermarket. *openai/gpt-oss-120b*: “Exit: ‘checkout’ as verb meaning to leave [...] Cart: shopping cart leads to checkout. Pay: checkout involves paying. Customer: checkout is for customers.” This is not an isolated case; in the most-frequent-answers table in the Analysis document, CHECKOUT was already the second most common answer for this item (91 occurrences). Checkout is a more elegant answer than the reference solution.
- guillo_24 (idea): several models arrive at idea through the lightbulb/flash-of-genius association (lightbulb moment). This is exactly the kind of visual idiom the dataset is trying to reach. *nvidia/nemotron-nano-9b-v2:free*: “‘light bulb’ is a common phrase for an idea. So ‘idea’ could connect lamp (light bulb), construct (build an idea)...”
- guillo_03 (city): a small model, *liquid/lfm-2.5-2.6b:free*, explicitly self-criticizes the quality of its own answers — an almost humorous touch of meta-textual self-awareness: “But ‘BUILDING’ is a bit too obvious and doesn’t feel like the kind of clever puzzle answer usually expected. ‘City’? [...] It seems too generic. Word-association games usually have a more specific or clever connection.” The model knows it must be clever, not just correct — a detail worth exploring elsewhere, regarding the kind of heuristic learned during training on puzzles/riddles.

Can we now give an answer to RQ6 (Which model gives the best answer, according to the author’s judgement?) after further analysis? As stated previously, a mathematical or probabilistic answer is not viable — we must rely on a philosophical-linguistic judgement dictated by reason.

Ludwig Wittgenstein, in 1921, within his *Tractatus Logico-Philosophicus*, wrote the famous maxim: “The limits of my language mean the limits of my world.” If a person does not possess the words to express a thought, that thought cannot find space in their logical reality. Later, in the *Philosophical Investigations*, he reformulates this, specifying that the meaning of a word does not lie in a corresponding object, but in its concrete use within a shared form of life. The Guillotine game is a distillate of these social and cultural uses, jargon expressions, idioms, and metaphors rooted in the everyday experience of human beings. An LLM brilliantly simulates adherence to these language games thanks to the monumental mass of training data, but does not take part in any biological or social form of life. Is the semantic association produced by the model, then, a genuine understanding of the

linguistic game, or a statistical simulation so refined as to be indistinguishable from the original? The data from this study show the structural discrepancy in difficulty (weighted Kappa 0.122–0.325) and the instability of the CoT, suggesting that LLMs play the Guillotine at a semantic table profoundly different from the human one.

The Sapir-Whorf hypothesis examines Native American languages (such as Hopi) and holds that the absence of certain verb tenses or grammatical categories can influence the perception of fundamental concepts (such as space and time) compared to Western languages. The strong version of this hypothesis — a total determination of thought by language, including Whorf’s original claims about the Hopi conception of time — is today considered by contemporary linguistics to be excessive and largely unsupported. Solid empirical evidence remains, however, for a weak version, linguistic relativity, according to which language influences, in a measurable but non-deterministic way, certain cognitive processes. Regarding colour, Kay & Kempton (1984) showed how the presence or absence of distinct lexical terms for blue and green affects the speed and threshold with which speakers categorize those colours, without nonetheless preventing perceptual distinction. Regarding grammar, Boroditsky and colleagues (2003) show how the grammatical gender assigned to objects in different languages (German, Spanish) influences the properties speakers attribute to those same objects. A similar pattern has also been observed in LLMs (Wang et al. 2025). The authors note how the structure of the language a model is trained on may be internalized into reasoning patterns specific to that language, reflecting reasoning biases typical of the training language.

The words with the greatest difficulty were: Petrichor (9), PC (10), watching (18), supermarket (21), emotion (22), concept (23), and scraping (29).

We can say right away that watch is more appropriate than watching (guardare and guardando in the Italian translation), since the gerund conjugation could have been avoided, leaving just watch. A similar point applies to supermarket and store, which are in fact synonyms. Perhaps the word supermarket is more used in Europe and store in the USA, but in terms of meaning it changes little. The remaining words, however, are objectively more complex to identify.

Let us take the five remaining words and play again: what do these words have in common? Petrichor, PC, emotion, concept, and scraping.

The author’s reasoning (a kind of human CoT) does not arrive at a single word common to the five clues, but at something more evolved. We can start the brainstorming by identifying the words as abstract concepts, but this does not fully fit — PC is a concrete thing. We must move up and identify a higher-level concept of generalization, like moving from atom to molecule. Each of these words can be a generalization of several components, a sort of container word into which various contents can be placed.

- **Petrichor** is the special case among these four; here it is not a problem of level of abstraction but of lexical rarity. It describes a very specific and narrow physical phenomenon — the smell perceivable when rain falls on dry ground, recognizable only at the onset of the storm and not throughout its duration. This specificity, together with its low frequency of use, makes it difficult for the models to retrieve, not a container structure like PC, emotion, concept, or scraping.
- **PC** is the generalization of a series of hardware and software components.
- **Emotion** is an enormous container of emotions, spanning from primary to secondary, with every kind of nuance
- **Concept** encompasses numerous identifications, such as ideas, theorems, assumptions, and other variants.

- **Scraping**, or scrap (here too the gerund could have been avoided), is part of a search for something — a sort of extraction of raw information to be processed afterward. Even when a cat scratches (si gratta) itself, it is responding to a sensory stimulus coming from a system other than the epidermis.

To use a metaphor: standing in front of a person, one can clearly make out the details of the face; moving up to the first floor of a building and looking at the same person, the details vanish. Going up to the tenth floor, or rising further still in a helicopter, one progressively loses even the outline of the figure. This rise leads to a higher level of abstraction, at the cost of detail. In the hardest cases in the dataset (concept, emotion, scraping, and in part PC), the models stay anchored to the lowest floor, answering with the most specific and concrete word compatible with the clues (idea, stress, spider, computer) instead of rising to the level of generality the solution requires. Hypernym and hyponym are the two words ready to precisely describe this gap in level. The correct solution is often the hypernym of what the models identify, which stays anchored to a more specific hyponym. Petrichor is the special case within this group. As seen in the Zipf analysis, here it is not a problem of level of abstraction but of lexical rarity — it is the most specific possible word in the dataset, not the most general, and must therefore be treated separately from the other four.

The author engaged in free-form brainstorming conversations with a few recent models (such as *Qwen 3.8 27B* or *Gemma 4 27B*), deliberately open-weight so as not to have too high a level of knowledge and complexity. Some interesting considerations emerged, particularly on the two items most interesting in terms of difficulty:

- guillo_09: rain; arid; perfume; terrain; lift => petrichor
- guillo_10: CPU; RAM; GPU; HD; VRAM => PC

When item 9 was put to the model, the answers were quite varied, but did not come close at all to the actual word identified. The particularly troublesome word among the five given was lift. The term has multiple meanings in English, ranging from “elevator,” “to abolish,” “to steal,” “a ride/tow,” and others. The word “to raise” or “to rise” was considered, but not necessarily associated with the other four.

Analysing prompt 10, the models gave even correct answers such as “hardware” and “computer,” but here a further association was needed — the metasemantic one, about how the names of the various components were written, namely as acronyms. The answers were correct, but written in the wrong form. Hardware was not precisely correct — external items such as keyboard or mouse should have been included too. Computer is 99% correct; it lacks the acronym form to reach 100%.

For item 9, the greatest difficulty was not only the word “lift,” but also the scarcity of the word. The word “petrichor” is a rare word to use. It is rarely used in normal conversation and probably even more rarely in training data. Presumably even tokenization does not treat it as a whole word, but splits it into sub-tokens, which makes identification even more complex. Item 10 requires a second level of abstraction — not just identifying the components inside a computer, but identifying them as acronyms. This calls for a higher level of reasoning and attention than mere words.

When the items were prepared, the author’s approach was one of reverse engineering. Starting from the final word, the five clue words were identified. In this way, fairly quickly, the thirty items of the project were identified. An LLM was deliberately not used to create the items, in order to avoid words that were too obviously matching each other.

One of the ideas offered by a commercial LLM was the following answer:

Petricore

- Perfume: Petricore is the typical “smell of rain.”
- Stone: Derives from the Greek pétros (stone) and ichór (the blood-like fluid of demons/gods).
- Earth: It is released when rain falls on dry earth.
- Drop: Born from the impact of the first drops of water with the ground.
- Summer: It is a particularly intense smell during summer storms after long dry spells.

PC

- Desktop: Contrasted with portable models (desktop PC).
- Personal: It is the acronym for *Personal Computer*.
- Bank: Can indicate either RAM memory (a “bank” of RAM) or a workstation/desk (banco in Italian covers both).
- Gaming: A leading category of the sector, the famous “gaming PCs.”
- Mouse: The pointing peripheral essential for using the PC.

In a new conversation, the clues were provided. For item 9 it failed to identify the correct word even after five attempts (in one iteration it even ran a web search). Once given the answer, it correctly identified the connection. Item 10, on the other hand, provided exactly the correct word “computer,” and not “PC,” because — rightly — the words were spelled out in full.

With two items and using only one model, this exercise has only an anecdotal, not a scientific, value, but it did lend support to the hypothesis of not having a model create the clues. Furthermore, it confirms the rarity of the word “petrichor,” scarcely used even by large LLMs, while the acronym PC requires a second level of abstraction for correct resolution.

For an LLM, operating exclusively within a closed syntactic network of textual references, Searle’s “Chinese room” argument holds that meaning arises only from the statistical co-occurrence of words, without any true referential anchoring to the real world. A skeptical reading, not shared by everyone, is the “systems reply” — and its variants argue that understanding can emerge from the system as a whole, not from the syntactic manipulation of the single component alone. Regardless of which philosophical reading one adopts, the empirical fact remains solid: if a term is too rare, its distributional statistics in the training data are too weak to support reconstructing the associative network needed to connect the five clues. The case of petrichor confirms this on an empirical basis, regardless of how one chooses to settle the philosophical dispute about understanding.

Among the identified items, some LLM answers are markedly more fitting than the ones devised. We have already identified watch and watching, or supermarket and store. We have other clues confirming how the models provided a more valid answer.

Guillo_22 (emotion), where the models suggest stress. Assessing the word against the clues, with no further levels of abstraction, is more precise compared to the author’s solution. Similarly guillo_24 (concept), where the models suggest idea, which comes closer to the given clues. A formal review by the author would radically change only these two elements; the other cases discussed are to be revised only partially (watching/watch and supermarket/store) in their semantics, but not in meaning.

When a human being reads, they go from the first word of the sentence to the last; inside the black box of the human brain, the thought of the meaning of the sentence read takes shape. Conversely, when a human being must write a sentence, through the chain of thought (conscious or unconscious) they formulate the sentence, but it is then written one word at a time. The broader a human being’s

vocabulary, the greater the depth with which they can describe their own world, as well as their ability to describe their own inner state. Presumably, LLMs too (however much we insist on calling them statistical next-word calculators) use words to enrich their world, or their representation of the world, using the information provided during training. To cite Plato's cave metaphor, at this point in history, both humans and LLMs probably find themselves in two adjacent caves. At the centre, the world and its reality, invisible to both. From both sides, shapes are projected to be interpreted. Humans must interpret the shapes cast by AIs, while AIs must interpret those cast by humans. At the moment we do not yet have one or more tools able to precisely analyse the models' reasoning, just as we do not have an adequate tool for understanding human reasoning. Any proposal, at this moment, is only part of the explanation, and not even a very truthful one, since we have no way of confirming or disproving the inferences we arrive at.

The difference between humans and LLMs is not so distant; in neurophysiology we know how to extrapolate and trace the paths of an electrical impulse from the motor cortex to the movement of a muscle, but we still have not identified what happens when we receive a question and give an answer. There are adequate theories that explain this process quite well, but we have no certainty of their truth. Mechanistic interpretability, much like neurology, seeks to discover what the real internal mechanisms are by which a stimulus leads to a response.

Language models possess correct propositional knowledge (Knowing-That, or knowing that — it associates MOUSE with countryside, whiskers, Mickey, lab) and repeatedly display it in the chain of thought. However, they lack the procedural competence and the metacognitive control (Knowing-How, or knowing how) needed to govern the process of logical convergence, to stop the search for useless alternatives, and to validate the correct intuition. This dissociation suggests that adding mere computing capacity, or more tokens, does not automatically amount to acquiring genuine procedural rationality.

A single, unambiguous answer about the best-performing model across the entire spectrum analysed is not easy to give. The author admits to having preferences for certain models, also due to their daily use outside this experiment. Bias and personal preferences are present in this decision, and this must be said honestly. Taking this limitation into account, on the whole the model the author finds most convincing, considering the adequate answers and the few evident reasoning errors observed, is *Claude Opus 5*, naming it as the model with the most satisfying performance among those considered.

Conclusions

Returning to the six questions posed at the outset, the data collected allow an answer to each.

RQ1/RQ2: no model systematically matches the author’s solution. The best among the open-weight models (*stepfun/step-3.5-flash*, 71.1%, 95% CI [61.0, 79.5]) and the best among the commercial ones (*x-ai/grok-4.6*, 76.7%, 95% CI [59.1, 88.2]) remain far from perfection, and neither is statistically distinguishable from the other top models in its own group (McNemar+FDR: 0 pairs out of 21 and out of 28, respectively). Including hand-validated synonyms, extended correctness rises to as much as 80.0% (*bytedance-seed/seed-2-1-turbo*): the quantitative distance between models and the reference solution is therefore real but not huge, and is partly an artefact of the automatic metrics alone, not of the models’ behaviour.

RQ3: yes, in specific cases. On at least three items (*guillo_21*, *guillo_22*, *guillo_18*), the models’ convergent answer turned out, on manual review, to be more coherent or more natural than the solution originally chosen by the author. The full manual review (100% of rows) also recovered 468 valid synonyms that the automatic metrics alone would never have counted among the exact answers.

RQ4: temperature, within the tested range (0.0–0.7), has no significant aggregate effect (Cochran’s $Q = 0.558$, $p = 0.757$), and the models’ ranking stays stable across temperatures (Spearman 0.886–0.930). Nine models show individually significant sensitivity without correction for multiple comparisons — a signal of response flexibility rather than of task instability.

RQ5: model size correlates strongly with accuracy ($\rho = 0.756$ on verified total parameters, $\rho = 0.690$ on active parameters per token, $n = 45$), but it is not the only factor: *gpt-oss:20b*, with only 3.61B active parameters per token, remains among the best on the list, confirming that the MoE and routing architecture carries as much weight as raw scale.

RQ6: from a statistical standpoint, this question has no unambiguous answer. The same pairwise comparison rules out a clear-cut winner in RQ1/RQ2, and the case here is equivalent, but the question asks for the author’s qualitative judgement, not a ranking. On that basis, the best candidate is *Claude Opus 5*, for the quality of its process rather than the exactness of its result. Its reasoning path on *petrichor* (*guillo_09*), the dataset’s hardest item due to lexical rarity, proceeds by systematic elimination of alternatives all the way to the correct solution. It does not display the pathological patterns observed elsewhere in the sample, such as the unrelated-code injection of *starcoder2:15b-instruct*, the unresolved oscillation between MOUSE and BEAST of *deepseek-r1:14b*, or the cyclical self-doubt of *qwen3.5:9b* on an item already solved correctly. These cases share a common trait: the model possesses the correct propositional knowledge (it knows exactly the answer), but lacks the procedural control to stop on it (it does not know how to stop). Ryle’s distinction between Knowing-That and Knowing-How, discussed further above, describes this gap better than a generic appeal to reasoning error.

Beyond the individual questions, three cross-cutting results deserve to be highlighted in closing. First, lexical rarity is a plausible factor only for the isolated case of *petrichor*, not a general mechanism of difficulty in the dataset (Spearman $\rho = 0.246$, not significant). Common items such as *guillo_10* (PC) remain as unsolved as, or more unsolved than, *petrichor*, because they require a leap of abstraction (from component to acronym) on top of a rare word. Second, the introduction of explicit reasoning in reasoning models carries a computational cost correlated with the difficulty perceived by the model

itself, but also exposes new failure modes absent in non-reasoning models (oscillations, loops, truncations) that a simple accuracy score does not capture. Third, human perception and model perception of an item's difficulty are related but not equivalent constructs (weighted Kappa 0.122–0.325). Generation time tells a story of difficulty actually experienced by the model, different from the one assigned a priori by the author. The result invites treating difficulty labels declared in benchmarks with caution.

Limitations and Future Research

For the sake of scientific honesty, the following methodological limitations must be made explicit:

- The RAT was not administered, because if the model had been trained on that data, the answers would not be the product of reasoning or association, but the retrieval of memorized information.
- N=1 on the reference judgement (solution, difficulty, coherence): being the work of a single author, it may be disputed. Deeper observations on the AI LLM models' answers compared to human evaluation are needed.
- RQ4 does not test reproducibility at fixed parameters, only sensitivity to temperature
- Sampling parameters are not uniform across commercial models, due to structural constraints of the APIs/interfaces (the OpenAI API does not expose top_k)
- Commercial models have only a single run (not 3 temperatures): comparable with the open-weight ones only on RQ1/2/3/5/6, not on RQ4

Future directions for deeper investigation and a better understanding include:

- Using Mechanistic Interpretability (MI) and Explainable AI (XAI) on models suited to data extraction
- Using a wider range of temperatures, beyond 1
- Modifying the top_p parameter as a further variable
- Comparing the RAT's answers with those from these prompts, with possible revision based on the data obtained from this initial experiment
- Submitting the items to human evaluation with $N > 30$ for a statistically comparable result

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Appendix I

prompt_id	category	word1	word2	word3	word4	word5	solution	difficulty	meaning
guillo_01	guillotine	gluteus	adjustable	sit	office	kitchen	chair	easy	gluteus is the muscle supported, the chair is adjustable in height, the chair is in the office or in the kitchen
guillo_02	guillotine	highway	wheel	transport	fuel	insurance	car	easy	the car can ride in highway, have a wheel, and is used to transport something, need fuel to run and insurance is mandatory for a car
guillo_03	guillotine	Washington	skyscraper	sewer	courthouse	shop	city	medium	Washington is a city name, skyscraper is in the city, sewer to drain the dirty water, courthouse is in every city and shop in the city
guillo_04	guillotine	Pascal	inch	stress	cloud	blood	pressure	medium	Pascal is the unit of pressure, inch in PSI that indicate pressure for inch square, pressure of material create stress, cloud is formed from low pressure, the blood pressure
guillo_05	guillotine	moon	speed	lamp	hope	visible	light	medium	moon light, speed of the light, light of the lamp, light of the hope, visible light
guillo_06	guillotine	videogame	human	society	advance	history	civilization	medium	Civilization is a videogame, human civilization, civilized society, advance in civilization, history of the civilization
guillo_07	guillotine	Patrick	sun	wars	five	multiplication	star	medium	Patrick star character of SpongeBob, the sun is a star, Star Wars the movie, a five-star hotel, star or asterisk is used to indicate the multiplication
guillo_08	guillotine	card	dental	suspension	nose	burn	bridge	hard	Use card in bridge game, dental bridge replaces teeth, suspension bridge structure, bridge of the nose, “burn your bridges” idiom
guillo_09	guillotine	rain	arid	perfume	terrain	lift	petrichor	hard	when rain in arid terrain the perfume lifted from terrain is called petrichor
guillo_10	guillotine	CPU	RAM	GPU	HD	VRAM	PC	easy	the acronym of a component of a PC (used acronym)
guillo_11	guillotine	language	snake	colt	monty	biceps	python	medium	Python is a programming language, python is a snake, colt python 357 magnum, Monty Python, slang in body builder for biceps
guillo_12	guillotine	qualification	book	academic	noble	job	title	medium	qualification title, academic title, noble title, job title
guillo_13	guillotine	gestational	politics	court	another	substitute	surrogate	hard	gestational surrogate, a surrogate politics is a man that talk for another, another people is a surrogate of me, a substitute of me is a surrogate
guillo_14	guillotine	geometric	meditation	art	powder	sanskrit	mandala	hard	Mandala is a geometric figure, used for meditation, and a type of art using a powder to crate it and is a Sanskrit word
guillo_15	guillotine	mean	math	analysis	descriptive	inferential	statistic	medium	mean is a math analysis to descript inferential conclusion in statistics
guillo_16	guillotine	aircraft	geometry	woodworking	thought	surface	plane	hard	in aviation aircraft is a plane, e geometric figure in two dimensions, in woodworking is a tool to flatten surface, in spiritual is a higher plane if thought, surface is a part of a cube
guillo_17	guillotine	cloth	rectangular	pole	symbol	signal	flag	medium	The flag is a piece of cloth generally rectangular fix in a pole used like a symbol of a nation or a signal to signaling something
guillo_18	guillotine	game	television	bird	observe	vigilante	watching	medium	watching a sport game, watching television, bird watching, observing is an action to watch, vigilante watching the residential zone
guillo_19	guillotine	double	red	fingers	angry	necklace	cross	hard	double-cross means betray, Red Cross organization, cross your fingers for luck, cross is British slang for angry, a cross pendant on a necklace
guillo_20	guillotine	exotic	adventure	fly	culture	relax	travel	medium	travel in a exotic location, adventure travel, travel using plane and fly, travelling to know different culture, relaxing travel
guillo_21	guillotine	exit	shelves	cart	pay	customer	supermarket	medium	everything is inside a supermarket
guillo_22	guillotine	reaction	response	driven	cognitive	adrenaline	emotion	medium	An emotion reaches a reaction and a response, emotion drives our behavior, is a cognitive study, emotion ca activates the adrenalin system
guillo_23	guillotine	philosophy	math	physics	business	society	concept	hard	Philosophy concept, math concept, physics concept, business concept, society

									concept
guillo_24	guillotine	construct	lamp	problem	innovation	good	idea	hard	idea is a construct, the lamp over the head when someone has an idea, idea that innove particular filed, a good idea
guillo_25	guillotine	tree	pie	computer	day	York	apple	easy	apple tree, apple pie dessert, Apple computer brand, “an apple a day” proverb, New York = the Big Apple
guillo_26	guillotine	love	Plato	essentialism	taoism	knowledge	philosophy	medium	Philosophy is the love for knowledge, essentialism is a part of philosophy, Plato as a philosopher, Taoism is a philosophy (and a religion)
guillo_27	guillotine	renewable	resource	environmental	social	economic	sustainability	hard	renewable energy is sustainability, sustainability resource, sustainability environmental, social sustainability, economic sustainability
guillo_28	guillotine	countryside	hole	whiskers	Mickey	lab	mouse	medium	mouse live in countryside in a hole, have a whisker, mickey mouse, laboratory mouse
guillo_29	guillotine	barrel	web	wood	rusty	cat	scraping	hard	scrape the bottom of the barrel, web scraping, wood scraping to abrading surface, scrape the rust from iron, cat scraping itself to clean your fur
guillo_30	guillotine	today	simple	gift	time	now	present	medium	Today is in present time, simple present a conjugation, present as a gift, present is a time, now is in the present

Appendix II

The Python files for the project are available at the following GitHub repository:
<https://github.com/AEONPHYTON/guillotine>